

ProveIt research continuation

Coincident Stieltjes Products and Directional Zeta Identities

Cubic finite parts, distributional contact terms,
coordinate invariance, and exact spectral reductions

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Abstract

This article continues the polylogarithm manuscript and six incoming research archives at a pinned ProveIt revision. We prove a finite holomorphic completion rule for coincident products of arbitrary Stieltjes argument derivatives, give its explicit cubic Tornheim realization, and reduce the requested cubic digamma finite part to the third derivative of the diagonal Tornheim function. The same local calculation evaluates its second derivative and, together with a Mellin decomposition, yields exact quadratic derivatives on every admissible asymmetric ray through the spectral origin.

We also determine all delta-supported corrections between periodic distributional correlations and canonical finite parts in the shift variable. A closed Gamma and harmonic-polynomial generator gives every Stieltjes index and every argument-derivative order. We classify all finite logarithmic singular germs invariant under coordinates with a prescribed identity jet, including necessity and sharp Stieltjes thresholds. For colored weighted sums, finite rational generating functions give all outer-order derivatives in ordinary polylogarithms, with exact Gamma and generalized-Stieltjes specializations. Finally, we give all principal Laurent coefficients and finite parts at a common-shift harmonic resonance with two independently moving exponents and arbitrary trailing depth.

Proofs specify the cutoff coordinate, the meromorphic restriction, and every allowed derivative direction. Classical ingredients and inherited results are credited; no general historical-priority claim is made. The Gaussian S_6 and current S_8 candidates remain conjectural. Reproducible exact and high-precision checks, narrowly scoped source corrections, and a proposed research agenda accompany the article.

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1 Scope, results, and conventions

1.1 The research questions being continued

The source corpus is the manuscript in `Analysis/Polylogarithms/docs/manuscript` and the six research archives in `docs/incoming`, frozen at the revision on the title page [1]. A read-only extraction verified the Git blob identity of 704 relevant text, code, data, bibliography, and archive files. This is a source-integrity statement, not a claim to have verified every theorem in those files. The mathematical audit is targeted to the questions and premises used below.

Three developments in the incoming material make a further exact calculus possible. First, periodic Stieltjes functions have been given consistent distributional and pointwise finite-part conventions. Second, nonlinear coordinate changes have been reduced to finite residue expressions. Third, the shifted height-one harmonic family has an exact Gamma-ratio representation with a finite Stirling specialization. The present work uses those developments as established inputs and extends the questions they explicitly leave open [2, 3, 4].

The main outcomes and their precise scope are as follows.

Incoming question or target	Result established here	Location
Coincident products beyond degree two	A finite local completion for every number of factors; a fully explicit cubic global kernel at all argument orders	Theorem 2.1; Proposition 2.2
Cubic digamma finite part	An exact reduction to $\omega'''(0)$, where $\omega(t) = T(t, t, t)$; also an exact evaluation of $\omega''(0)$	Theorem 2.3
Contact terms at a shift collision	One explicit Gamma-harmonic generator for every delta coefficient; no undetermined lower delta derivatives	Theorem 3.1
Coordinate-invariant singular germs	Necessary and sufficient coefficient conditions for every prescribed order of tangency	Theorems 4.1, 4.3
Normal Tornheim derivatives at integer inner orders	Finite depth-two reductions at all integer inner orders, and depth-one closure at nonpositive orders with arbitrary weights	Theorems 5.1, 5.3
Spectral direction dependence	Constant, linear, and quadratic terms on every ray with $(a + c)(b + c) \neq 0$	Theorem 6.1
Independent harmonic exponents	Complete Laurent principal parts and finite parts for two moving exponents at every trailing depth and common shift	Theorem 7.3

These statements answer several archived research questions, but they do not all have the same kind of closure. The cubic finite part is reduced to a specified multizeta derivative, not evaluated in ordinary zeta constants alone. The positive-inner-order colored formulas close in depth-two polylogarithms. The weighted nonpositive-inner-order theorem closes in ordinary polylogarithms and their order derivatives. The two-exponent harmonic theorem eliminates its holomorphic remainder through the finite part, but does not eliminate the remainder from every higher regular coefficient.

1.2 Relation to the literature

The analytic ingredients are classical. Hurwitz Fourier expansions and product integrals are treated by Espinosa and Moll [6, 7]. Bailey and Borwein systematically study Mordell–Tornheim

order derivatives and their polylogarithmic integral representations [8]. Colored Tornheim partial fractions already appear in Zhao [13]. Borwein and Dilcher give direction-dependent Tornheim values and first derivatives on the equal-inner-order family [9]; related Witten values are discussed by Romik [14]. Logarithmic finite parts and their change of variables have an established theory [18]. Meromorphic continuation and Laurent algorithms for multiple zeta functions at integer points are supplied by Akiyama–Egami–Tanigawa and Matsumoto–Onozuka–Wakabayashi [15, 16].

Accordingly, this article makes no claim to have discovered finite parts, the Tornheim–Hurwitz relation, directional regularization, or multiple-zeta continuation. The contributions are the stated explicit completion and contact generators, the invariant-subspace classification, the asymmetric quadratic formula, and the finite two-exponent harmonic coefficient rule relative to the audited corpus. The elementary weighted identities are included as proved, useful consequences of a general rational coefficient method.

1.3 Notation and finite-part prescriptions

We use

$$\zeta(1+u, a) = \frac{1}{u} + \sum_{n \geq 0} \frac{(-1)^n}{n!} \gamma_n(a) u^n, \quad \gamma_0(a) = -\psi(a), \quad \gamma_n = \gamma_n(1). \quad (1.1)$$

Here $\psi = \Gamma'/\Gamma$, and $\gamma = \gamma_0$ is Euler’s constant. A prime on ψ is an argument derivative. On $\zeta(s, a)$, a prime or $[k]$ superscript denotes a derivative in s unless an argument derivative is explicitly written. Thus $\zeta^{[k]}(s, a) = \partial_s^k \zeta(s, a)$ and $\text{Li}_s^{[k]}(z) = \partial_s^k \text{Li}_s(z)$. All integer summation bases are raised to complex powers using their real logarithms. A common shift a is initially in $\Re a > 0$, with the analytic logarithm in that half-plane.

The generalized harmonic numbers are

$$H_N^{(j)} = \sum_{\nu=1}^N \nu^{-j}, \quad H_N = H_N^{(1)}, \quad H_0 = 0.$$

The rising factorial is $(a)_n = a(a+1) \cdots (a+n-1)$; the falling factorial is $a^{\underline{n}} = a(a-1) \cdots (a-n+1)$. The second-kind Stirling number is $\{n \atop k\}$; $s(n, k)$ denotes the signed first-kind Stirling number. Coefficient extraction $[u^j]F$ always refers to an ordinary power series, with any exponential-generating factorials displayed separately.

For a function with a finite logarithmic singular expansion at zero,

$$\text{FP} \int_0^1 f(x) dx = \text{CT}_{\varepsilon \downarrow 0} \int_\varepsilon^1 f(x) dx. \quad (1.2)$$

The constant term means the coefficient of $\varepsilon^0(\log \varepsilon)^0$ after discarding divergent powers and logarithms. It is a coordinate-dependent prescription. A spectral Laurent constant $\text{CT}_{u=0} F(u)$ is a different operation; identifying the two requires a proof of their local counterterms. Such a proof is given in Section 2.

For periodic distributions, δ is the unit Dirac mass at the origin of $\mathbb{R}/\mathbb{Z}$, D is the distribution derivative, and $\check{F}(x) = F(-x)$. We normalize convolution so that

$$(\check{F} * G)(a) = \int_0^1 F(x) G(x+a) dx$$

whenever ordinary integration is valid. A finite part of a pointwise product is not defined by multiplying distributions. Section 3 explicitly compares two independently defined extensions of the same off-collision function.

Why the directions matter. There is no single holomorphic Taylor germ of the uncolored Tornheim function at $(0, 0, 0)$. For example,

$$T(t, t, t)|_{t=0}^{\text{cont}} = \frac{1}{3}, \quad T(0, 0, t)|_{t=0}^{\text{cont}} = \frac{5}{12}.$$

A notation such as $T''(0)$ without a specified one-variable restriction is therefore insufficient. The same distinction occurs when one exponent of a harmonic sum is fixed before differentiating another. Every derivative formula below preserves its stated restriction.

2 Coincident products: an explicit holomorphic completion

The newest coincident-point report evaluates all quadratic Stieltjes moments and explicitly asks for the cubic case, beginning with FP $\int_0^1 \psi(x)^3 dx$ [2]. At three factors the local subtraction has a new feature: a pair of singular factors and the linear term of the remaining factor together produce a logarithmic divergence. We give its exact spectral counterterm. The construction works at every derivative order and, locally, at every number of factors.

2.1 A finite local rule for any number of factors

Use the lowering spectral convention

$$g(\lambda, x) = \zeta(1 - \lambda, x) + \frac{1}{\lambda} = \sum_{m \geq 0} \gamma_m(x) \frac{\lambda^m}{m!}. \quad (2.1)$$

The pole at $\lambda = 0$ is removable at each $x > 0$. For $r \geq 0$ put

$$c_r(\lambda) = (-1)^r (1 - \lambda)_r, \quad h_r(\lambda, x) = \partial_x^r g(\lambda, 1 + x).$$

The Hurwitz shift relation gives the exact splitting

$$\partial_x^r g(\lambda, x) = c_r(\lambda) x^{\lambda-r-1} + h_r(\lambda, x). \quad (2.2)$$

Every h_r is holomorphic jointly in λ near zero and x near zero. Its required Taylor coefficients are explicit:

$$[x^k] h_r(\lambda, x) = \begin{cases} \zeta(1 - \lambda) + 1/\lambda, & r = k = 0, \\ c_{r+k}(\lambda) \zeta(1 + r + k - \lambda)/k!, & r + k > 0. \end{cases} \quad (2.3)$$

Let $d \geq 2$ and $\mathbf{r} = (r_1, \dots, r_d)$ be nonnegative integers. For $\emptyset \neq S \subseteq \{1, \dots, d\}$ write

$$\lambda_S = \sum_{i \in S} \lambda_i, \quad \nu_S = \sum_{i \in S} (r_i + 1) - 1,$$

and define the finite numerator

$$A_S(\boldsymbol{\lambda}) = \left(\prod_{i \in S} c_{r_i}(\lambda_i) \right) [x^{\nu_S}] \prod_{j \notin S} h_{r_j}(\lambda_j, x). \quad (2.4)$$

The empty smooth product is 1. In particular, $A_{\{1, \dots, d\}} = 0$ because $\nu_{\{1, \dots, d\}} \geq 1$. Let $\mathcal{L}_{\mathbf{r}}$ denote the meromorphic continuation of the ordinary integral of $\prod_i \partial_x^{r_i} g(\lambda_i, x)$. It is initially defined, for example, with every $\Re \lambda_i > r_i + 1$, away from the removable parameter expressions.

Theorem 2.1 (Subset completion in the unit coordinate). *The expression*

$$\mathcal{F}_r(\lambda) = \mathcal{L}_r(\lambda) - \sum_{\emptyset \neq S \subseteq \{1, \dots, d\}} \frac{A_S(\lambda)}{\lambda_S} \quad (2.5)$$

is holomorphic in a neighborhood of the spectral origin. Its Taylor coefficients are exactly

$$\text{FP} \int_0^1 \prod_{i=1}^d \gamma_{m_i}^{(r_i)}(x) dx = \left(\prod_i m_i! \right) [\lambda_1^{m_1} \cdots \lambda_d^{m_d}] \mathcal{F}_r(\lambda). \quad (2.6)$$

Here FP is the constant term of the cutoff integral in the fixed coordinate x , with upper endpoint 1 and no additional finite term. The complete numerators in (2.5) are subtracted; restricting a numerator to its polar hyperplane generally changes the finite answer.

Proof. Fix $0 < b < 1$. Expanding the exact product in (2.2) gives one term for each singular subset S , namely

$$\left(\prod_{i \in S} c_{r_i}(\lambda_i) \right) x^{\lambda_S - \nu_S - 1} \prod_{j \notin S} h_{r_j}(\lambda_j, x).$$

For each nonempty S , subtract the Taylor polynomial of its smooth product through degree ν_S . The remainder is normally integrable on $(0, b)$ for all parameters in a sufficiently small common polydisc; fixed parameter derivatives only add powers of $\log x$. The Taylor monomial of degree k has integral proportional to

$$\frac{b^{\lambda_S + k - \nu_S}}{\lambda_S + k - \nu_S}.$$

Only $k = \nu_S$ has a denominator vanishing at the spectral origin. Its meromorphic integral is $A_S b^{\lambda_S} / \lambda_S$. The generating function of its coordinate finite parts is $A_S(b^{\lambda_S} - 1) / \lambda_S$. Their difference is precisely A_S / λ_S . All other denominators are nonzero near the origin and their Taylor coefficients equal the corresponding cutoff constant terms. The integral on $[b, 1]$ is holomorphic and has the same coefficient-extraction property. Summing the finitely many subsets proves both assertions. Changing b redistributes ordinary integrals and Taylor integrals equally on both sides, so the result does not depend on this auxiliary split. $\square$

This theorem is a local completion rule, not a claim that an arbitrary depth integral reduces to single zeta values. For three factors its global coordinates are the classical Tornheim function and ordinary zeta functions. That explicit global formula comes next.

2.2 The cubic global kernel and all its argument derivatives

Define, initially in its domain of absolute convergence,

$$T(A, B, C) = \sum_{m, n \geq 1} \frac{1}{m^A n^B (m+n)^C}. \quad (2.7)$$

We always use its meromorphic continuation, with the order of any restriction specified. The classical Hurwitz–Tornheim connection is due to earlier work, including Espinosa and Moll [7]; the contribution here is its complete coordinate normalization at the coincident Stieltjes point.

Put $\Lambda = \lambda_1 + \lambda_2 + \lambda_3$. If $\{i, j, k\}$ is $\{1, 2, 3\}$, define

$$K_2(a, b) = \frac{2\Gamma(a)\Gamma(b)}{(2\pi)^{a+b}} \cos \frac{\pi(a-b)}{2} \zeta(a+b), \quad (2.8)$$

$$K_3(\lambda) = \frac{2 \prod_i \Gamma(\lambda_i)}{(2\pi)^\Lambda} \sum_{k=1}^3 \cos \frac{\pi(\lambda_i + \lambda_j - \lambda_k)}{2} T(\lambda_i, \lambda_j, \lambda_k). \quad (2.9)$$

In (2.9), the two indices other than k may be taken in increasing order, since $T(A, B, C) = T(B, A, C)$.

Proposition 2.2 (Finite spectral expression). *The kernels K_2 and K_3 are the meromorphic continuations of the integrals of two and three factors $\zeta(1 - \lambda_i, x)$, respectively. For $d = 3$, the meromorphic integral in Theorem 2.1 is explicitly*

$$\mathcal{L}_r(\lambda) = \left(\prod_i c_{r_i}(\lambda_i) \right) \left\{ K_3(\lambda_1 - r_1, \lambda_2 - r_2, \lambda_3 - r_3) + \sum_{k:r_k=0} \frac{K_2(\lambda_i - r_i, \lambda_j - r_j)}{\lambda_k} + \frac{\mathbf{1}_{r=0}}{\lambda_1 \lambda_2 \lambda_3} \right\}. \quad (2.10)$$

Together with (2.4), this is a finite Tornheim–zeta coefficient formula for every cubic Stieltjes moment, at arbitrary Stieltjes and argument-derivative orders.

Proof. For $\Re \lambda_i > 1$, the nonzero Fourier coefficient of $\zeta(1 - \lambda_i, x)$ at n is

$$\Gamma(\lambda_i)(2\pi|n|)^{-\lambda_i} e^{-\pi i \lambda_i \operatorname{sgn}(n)/2}.$$

The Fourier series are absolutely convergent. Integrating their product keeps $n_1 + n_2 + n_3 = 0$, with all $n_i \neq 0$. There are three choices of the negative frequency and their three sign reversals. On each positive cone the negative magnitude is $m + n$, giving exactly (2.9). The two-factor calculation gives (2.8); it is the classical Hurwitz product formula [6]. Finite endpoint Taylor subtraction supplies meromorphic continuation of the original integrals, and uniqueness extends these identities. Finally, $\partial_x^r g = c_r \zeta(1 + r - \lambda, x)$ for $r > 0$, whereas $g = \zeta(1 - \lambda, x) + 1/\lambda$ for $r = 0$. Expand the three factors. Integrals with just one Hurwitz factor vanish by meromorphic continuation of $\int_0^1 \zeta(s, x) dx = 0$. The other terms give (2.10). $\square$

For undifferentiated factors set $h(\lambda) = \zeta(1 - \lambda) + 1/\lambda$. Formula (2.5) becomes particularly short:

$$\begin{aligned} \mathcal{F}_0 = K_3 &+ \sum_{k=1}^3 \frac{K_2(\lambda_i, \lambda_j)}{\lambda_k} + \frac{1}{\lambda_1 \lambda_2 \lambda_3} - \sum_{i=1}^3 \frac{h(\lambda_j)h(\lambda_k)}{\lambda_i} \\ &+ \sum_{k=1}^3 \frac{(1 - \lambda_k)\zeta(2 - \lambda_k)}{\lambda_i + \lambda_j}. \end{aligned} \quad (2.11)$$

The last line is necessary. It is the pair-subset correction absent from the separated-singularity triple formula. Neither it nor the singleton correction can be suppressed before restricting spectral directions.

2.3 The stated cubic digamma target

Let

$$\omega(t) = T(t, t, t), \quad L = \log(2\pi), \quad E(t) = \Gamma(1 + t)e^{-Lt}, \quad \gamma = \gamma_0(1).$$

The symbol $\omega^{(j)}(0)$ means a derivative of this specified one-variable continuation. It does not mean a derivative of a jointly holomorphic T at $(0, 0, 0)$.

Theorem 2.3 (Diagonal second derivative and cubic digamma moment). *The diagonal continuation is holomorphic at zero and satisfies*

$$\omega(0) = \frac{1}{3}, \quad \omega'(0) = L, \quad \boxed{\omega''(0) = L^2 - 4\zeta''(0)}. \quad (2.12)$$

The first two values are classical [9]. The coordinate-normalized cubic moment is

$$\begin{aligned} \boxed{\operatorname{FP} \int_0^1 \psi(x)^3 dx} &= -\omega'''(0) - 8\zeta'''(0) - 12(\gamma + L)\zeta''(0) - L^3 - 6\gamma L^2 \\ &\quad + 5\gamma^3 + 6\gamma\gamma_1 - \frac{3}{2}\gamma\zeta(2) + \frac{3}{2}(\zeta(2) + \zeta'(2)). \end{aligned} \quad (2.13)$$

Thus the requested cubic finite part is exactly reduced to one third derivative of the diagonal Tornheim function and ordinary zeta and Stieltjes constants. No reduction of $\omega'''(0)$ to ordinary zeta constants is asserted.

Proof. On the diagonal, (2.11) reads

$$F(t) := \mathcal{F}_0(t, t, t) = \frac{6E(t)^3 \cos(\pi t/2) \omega(t) + 6E(t)^2 \zeta(2t) + 1}{t^3} - \frac{3h(t)^2}{t} + \frac{3(1-t)\zeta(2-t)}{2t}. \quad (2.14)$$

The left side is holomorphic by Theorem 2.1. Solving this identity for $\omega(t)$ proves its holomorphy at zero, since $E(0) = 1$ and $\cos 0 = 1$. Write

$$E(t) = \exp \left[-(\gamma + L)t + \frac{\zeta(2)}{2}t^2 - \frac{\zeta(3)}{3}t^3 + O(t^4) \right].$$

Cancellation of the coefficients of t^{-3} and t^{-2} gives the first two values of (2.12), using $\zeta(0) = -1/2$ and $\zeta'(0) = -L/2$. The numerator coefficient at t^2 is

$$3\omega''(0) + 3\gamma^2 - 3L^2 - \frac{3}{2}\zeta(2) + 12\zeta''(0).$$

The other two terms of (2.14) contribute $-3\gamma^2 + 3\zeta(2)/2$ to the coefficient of t^{-1} . This proves the second-derivative formula.

The numerator coefficient at t^3 , after those substitutions, is

$$\omega'''(0) + 8\zeta'''(0) + 12(\gamma + L)\zeta''(0) + L^3 + 6\gamma L^2 - 5\gamma^3 + \frac{3}{2}\gamma\zeta(2).$$

The remaining constant terms are $-6\gamma\gamma_1 - 3(\zeta(2) + \zeta'(2))/2$. Finally $F(0) = \text{FP} \int \gamma_0(x)^3 dx = -\text{FP} \int \psi(x)^3 dx$, which proves (2.13). $\square$

A direct, convergent integral gives an independent way to evaluate the same left side. The expansion $\psi(x) = -x^{-1} - \gamma + \zeta(2)x - \zeta(3)x^2 + \dots$ gives

$$\text{FP} \int_0^1 \psi(x)^3 dx = \int_0^1 \left[\psi(x)^3 + \frac{1}{x^3} + \frac{3\gamma}{x^2} - \frac{3\zeta(2) - 3\gamma^2}{x} \right] dx + \frac{1}{2} + 3\gamma. \quad (2.15)$$

The integrand has a finite limit at zero. Independent truncated Mellin-series evaluation of $\omega'''(0)$ and local-series quadrature of (2.15) agree to more than thirty decimal places:

$$\begin{aligned} \omega''(0) &= 11.403217934867761113621595961662\dots, \\ \omega'''(0) &= 86.657866601100073652903276858762\dots, \\ \text{FP} \int_0^1 \psi(x)^3 dx &= 1.966262734391534340664375338411\dots \end{aligned} \quad (2.16)$$

These are numerical diagnostics accompanying the analytic proof; the implementation does not claim interval-certified final digits.

Corollary 2.4 (A cubic derivative specialization). *In the same coordinate convention,*

$$\text{FP} \int_0^1 \psi(x)^2 \psi'(x) dx = \zeta(3) - 2\gamma\zeta(2). \quad (2.17)$$

Proof. Take the constant term of $\int_\epsilon^1 \psi^2 \psi' = (\psi(1)^3 - \psi(\epsilon)^3)/3$. The constant coefficient of $\psi(\epsilon)^3$ is $-\gamma^3 + 6\gamma\zeta(2) - 3\zeta(3)$. $\square$

More generally, a finite-part integral of a derivative of a product equals its upper-endpoint value minus the constant term of its local expansion. This supplies a terminating family of relations between the cubic moments in (2.6); it does not impose a spurious independent value on the remaining primitive cubic moment.

2.4 Why a spectral direction must be recorded

The reduction at fixed $A = B = 0$ gives

$$T(0, 0, C) = \zeta(C - 1) - \zeta(C). \quad (2.18)$$

Indeed, $m + n = N$ has $N - 1$ positive solutions. Continuing this one-variable identity gives $5/12$ at $C = 0$, whereas (2.12) gives $1/3$ on the diagonal. This direction dependence is already explicit in Borwein and Dilcher [9, Theorem 2.7]. It is not an inconsistency. The multivariable function is meromorphic at intersecting polar hyperplanes, and its restrictions carry different finite information. The completed function $\mathcal{F}_r$ is genuinely holomorphic; the uncompleted T generally is not. Section 6 computes the first two derivatives along an arbitrary transverse ray.

3 The full distributional collision correction

The incoming coincident-point report isolates the off-point collision singularity and its finite constant, but leaves the distribution supported at the collision undetermined [2]. We now specify the extension convention and calculate that distribution for every pair of Stieltjes indices and every pair of argument derivatives. This is a comparison of two extensions of the same off-point function. It does not identify a pointwise product of two singular distributions at the same argument.

3.1 The two extensions being compared

Work on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. Let G_m be the unit-coordinate Hadamard extension of $\gamma_m(x)$ from $0 < x < 1$, and let $F_{m,p}$ be the corresponding extension of $\gamma_m^{(p)}(x)$. Thus, for a smooth periodic test function,

$$\langle F_{m,p}, \phi \rangle = \text{CT}_{\epsilon \downarrow 0} \int_{\epsilon}^1 \gamma_m^{(p)}(x) \phi(x) dx, \quad F_{m,0} = G_m. \quad (3.1)$$

Write $D = d/dx$, $\Delta = \delta_0 - 1$, and $\check{U}(x) = U(-x)$. The exact correlation is the well-defined periodic convolution

$$\mathcal{R}_{mn}^{p,q} = \check{F}_{m,p} * F_{n,q}. \quad (3.2)$$

For ordinary smooth functions this is $\int_0^1 f(x)g(x+a) dx$ as a function of a .

For $0 < a < 1$, let $J_{mn}^{p,q}(a)$ be the unit-coordinate finite part of the separated product $\int_0^1 \gamma_m^{(p)}(x) \gamma_n^{(q)}(\{x+a\}) dx$. It is the off-point restriction of (3.2), as established in the manuscript's translated contact law [1]. Define its canonical extension in the shift coordinate by

$$\langle \mathfrak{F}_{mn}^{p,q}, \phi \rangle = \text{CT}_{\epsilon, \eta \downarrow 0} \int_{\epsilon}^{1-\eta} J_{mn}^{p,q}(a) \phi(a) da. \quad (3.3)$$

The two endpoint constants are taken independently in a and $1 - a$; there is no added delta term. The local power-log expansions proved in the incoming collision report guarantee existence.

3.2 Finite generating polynomials

This section uses a raising parameter to simplify the Gamma quotient:

$$\mathcal{G}(u) = \sum_{m \geq 0} (-1)^m G_m \frac{u^m}{m!}, \quad \mathcal{F}_p(u) = \sum_{m \geq 0} (-1)^m F_{m,p} \frac{u^m}{m!}.$$

For $r \geq 0$ set

$$P_r(z) = \prod_{j=1}^r (1 + z/j), \quad H_r(z) = \frac{P_r(z) - 1}{z}, \quad P_0 = 1, \quad H_0 = 0. \quad (3.4)$$

H_r is a polynomial, not the scalar harmonic number; its constant coefficient is $\sum_{j=1}^r 1/j$. All its coefficients are elementary symmetric polynomials in $1, 1/2, \dots, 1/r$.

Put $w = u + v$ and define

$$A(u, v) = \frac{\Gamma(1-u)\Gamma(1+w)}{\Gamma(1+v)}, \quad T_0(u, v) = \frac{A(u, v) - 1}{uw}, \quad B(u, v) = T_0(u, v) + T_0(v, u). \quad (3.5)$$

The quotients are removable because $A(0, v) = A(u, -u) = 1$. The convergent local series

$$\log A(u, v) = \sum_{k \geq 2} \frac{\zeta(k)}{k} \left[u^k + (-1)^k ((u+v)^k - v^k) \right] \quad (3.6)$$

makes every coefficient an exact finite polynomial in positive integer zeta values. The undifferentiated off-point use of this Gamma quotient is inherited from the manuscript [1].

There is also a compact symmetric form:

$$B(u, v) = \frac{1}{uv} \left\{ \frac{\Gamma(1-u)\Gamma(1-v)}{\Gamma(1-u-v)} \frac{\cos(\pi(u-v)/2)}{\cos(\pi(u+v)/2)} - 1 \right\}. \quad (3.7)$$

Indeed, $uvB = (vA(u, v) + uA(v, u))/(u+v) - 1$. The Gamma reflection formula turns the fraction before the subtraction into the Gamma quotient in (3.7) times $\{\sin(\pi u) + \sin(\pi v)\}/\sin(\pi(u+v))$. The sine addition formula gives the displayed cosine ratio. All apparent quotients are interpreted by their removable holomorphic extensions near $(0, 0)$. This second form provides an independent coefficient check on the generator.

Theorem 3.1 (All-index collision contact distribution). *Let $m, n, p, q \geq 0$ and $r = p + q$. Then*

$$\boxed{\mathcal{R}_{mn}^{p,q} = \mathfrak{F} J_{mn}^{p,q} + e_{mn}^{p,q} \delta_0^{(r)}}. \quad (3.8)$$

There are no additional lower derivatives of δ_0 . The coefficient is explicitly

$$e_{mn}^{p,q} = (-1)^{m+n} m!n! [u^m v^n] E_{p,q}(u, v), \quad (3.9)$$

where the following holomorphic germ requires only polynomial arithmetic and (3.6):

$$E_{p,q}(u, v) = (-1)^p \left\{ P_r(w) B(u, v) + \frac{H_r(w) - H_r(v)}{u} + \frac{H_r(w) - H_r(u)}{v} - H_p(u) H_r(v) - H_q(v) H_r(u) + H_p(u) H_q(v) \right\}. \quad (3.10)$$

Both divided differences in this formula are polynomials.

Proof. Let Z_s be the meromorphic periodic Hurwitz distribution. Its Fourier coefficients are the Hurwitz coefficients used in Section 2, with zero mean. Local Taylor subtraction at the origin gives

$$Z_{1+u} = -\frac{\Delta}{u} + \mathcal{G}(u). \quad (3.11)$$

For $p > 0$, the same subtraction at the higher Hurwitz pole, together with $D^p Z_s = (-1)^p (s)_p Z_{s+p}$, gives

$$\mathcal{F}_p(u) = D^p \mathcal{G}(u) + H_p(u) \delta_0^{(p)}. \quad (3.12)$$

This generating version of the manuscript's differentiation anomaly also holds for $p = 0$. For clarity, the polar term of Z_{1+p+u} is $(-1)^{p+1} \delta_0^{(p)}/(p!u)$. Multiplication by $(-1)^p (1+u)_p$ produces $-P_p(u) \delta_0^{(p)}/u$; subtracting the pole of the left side leaves exactly (3.12).

The beta form of the classical correlation, continued as a periodic distribution, is

$$\check{Z}_{1+u} * Z_{1+v} = -\frac{A(u, v)}{u} Z_{1+w} - \frac{A(v, u)}{v} \check{Z}_{1+w}.$$

It follows either from the Fourier coefficients or first in a convergent half-plane and then by meromorphic continuation. Substitute (3.11) and use $\Delta * \Delta = \Delta$ and $\Delta * G_m = G_m$. The result is the holomorphic distribution identity

$$\begin{aligned} \check{\mathcal{G}}(u) * \mathcal{G}(v) &= B(u, v)\Delta - \frac{\mathcal{G}(w) - \mathcal{G}(v)}{u} - \frac{\check{\mathcal{G}}(w) - \check{\mathcal{G}}(u)}{v} \\ &\quad - w[T_0(u, v)\mathcal{G}(w) + T_0(v, u)\check{\mathcal{G}}(w)]. \end{aligned} \quad (3.13)$$

Its off-point constant is $-B$. The canonical shift extension replaces each ordinary Stieltjes function by its corresponding G_m and retains that constant. Consequently the difference for $r = 0$ is $B\delta_0$, as claimed; it is not $B\Delta$.

For $r > 0$, converting $D^r \mathcal{G}(z)$ to the canonical extension of its ordinary derivative adds the difference $-H_r(z)\delta_0^{(r)}$. A reflected term has the same difference: reflection and the chain rule each contribute $(-1)^r$. Applying this to (3.13), the contact coefficient of its r th derivative is

$$P_r(w)B + \frac{H_r(w) - H_r(v)}{u} + \frac{H_r(w) - H_r(u)}{v}.$$

Finally expand the correlation of $D^p \mathcal{G}(u) + H_p(u)\delta_0^{(p)}$ and $D^q \mathcal{G}(v) + H_q(v)\delta_0^{(q)}$. The two cross terms and the delta-delta term supply, respectively, $-H_p(u)H_r(v)$, $-H_q(v)H_r(u)$, and $H_p(u)H_q(v)$. The orientation of the first factor contributes $(-1)^p$. This proves (3.10). Every contact term in this calculation has derivative order r , proving the absence of lower orders. Coefficient extraction gives (3.8). $\square$

3.3 Explicit contact coefficients

For undifferentiated factors, $E_{0,0} = B$ and the first values are

(m, n)	$e_{mn}^{0,0}$	contact term
$(0, 0)$	$2\zeta(2)$	$2\zeta(2)\delta_0$
$(1, 0)$ or $(0, 1)$	$\zeta(3)$	$\zeta(3)\delta_0$
$(2, 0)$ or $(0, 2)$	$11\zeta(4)/2$	$11\zeta(4)\delta_0/2$
$(1, 1)$	$7\zeta(4)/2$	$7\zeta(4)\delta_0/2$

The last two simplifications use $\zeta(2)^2 = 5\zeta(4)/2$. For Stieltjes indices $m = n = 0$ and positive argument derivatives,

$$\begin{aligned} e_{00}^{1,0} &= 1 - 2\zeta(2), \\ e_{00}^{1,1} &= 1 - 2\zeta(2), \\ e_{00}^{2,0} &= 2\zeta(2) - \frac{5}{4}. \end{aligned} \quad (3.14)$$

The associated distributions are δ'_0 , δ''_0 , and δ'''_0 , respectively. Swapping the two factors reflects the correlation; accordingly

$$e_{nm}^{q,p} = (-1)^r e_{mn}^{p,q}. \quad (3.15)$$

This identity is also immediate from (3.10).

For instance, the known off-point identity is $J_{00}^{0,0}(a) = \gamma_1(a) + \gamma_1(1-a) - 2\zeta(2)$. Its exact periodic distributional completion is

$$\check{G}_0 * G_0 = G_1 + \check{G}_1 + 2\zeta(2)(\delta_0 - 1). \quad (3.16)$$

An off-point numerical computation cannot see the delta term. In contrast, a nonzero Fourier coefficient of (3.16) contains the necessary constant $2\zeta(2)$; omitting it gives the wrong correlation even though the two functions agree for $0 < a < 1$.

3.4 Normalized primitives of the contact discrepancy

The contact formula also fixes constants in antiderivatives. Let $\mathcal{I}$ be the zero-mean inverse of D on periodic distributions of mean zero. For the periodic Bernoulli polynomial $B_j(\{a\})$,

$$\mathcal{I}^j \Delta = -\frac{B_j(\{a\})}{j!}, \quad j \geq 1. \quad (3.17)$$

This follows from $DB_1(\{a\}) = 1 - \delta_0$ and $B'_j = jB_{j-1}$ for $j > 1$, with zero means. For $r > 0$, applying $\mathcal{I}^{r+j}$ to the discrepancy in (3.8) yields

$$\mathcal{I}^{r+j}(\mathcal{R}_{mn}^{p,q} - \mathfrak{F}J_{mn}^{p,q}) = -e_{mn}^{p,q} \frac{B_j(\{a\})}{j!}, \quad j \geq 1. \quad (3.18)$$

For $r = 0$, first subtract the mean of the discrepancy, replacing $e\delta_0$ by $e\Delta$, and the same formula applies. These are fixed normalized primitives, with no unspecified additive polynomial. They explain exactly how a point-supported correction can become an ordinary Bernoulli term after repeated integration.

4 Which logarithmic singularities have no coordinate defect?

The nonlinear coordinate formulas in the two incoming reports give a finite algorithm for each specified singular germ. The fifth research question in the Exact Identities report asks for an intrinsic classification of the germs whose defect vanishes under every coordinate tangent to the identity. We answer that question, and more generally allow any prescribed order of tangency. The necessary condition is as important as the sufficient one: logarithmic degrees from different pole orders cannot conceal a defect for every coordinate.

The change-of-variables mechanism for logarithmic finite parts is classical; see, for example, Lätt's Theorem 2 in [18]. The theorem below is an exact invariant-subspace classification and a specialization to the Stieltjes hierarchy relative to the audited incoming material. No general historical-priority claim is made.

4.1 Conventions and the inherited residue formula

Work locally on the positive side of the origin. A logarithmic meromorphic germ has the unique singular expansion

$$h(x) = \sum_{q=1}^R \sum_{n=0}^L c_{q,n} x^{-q} \log^n x + h_{\text{reg}}(x), \quad (4.1)$$

where the omitted part is locally integrable and is a finite sum of analytic germs times nonnegative powers of $\log x$. Coefficients may be real or complex. Define $\text{FP}_x h$ by taking the constant term of its cutoff integral against a smooth test function. Positive powers of the cutoff logarithm and negative powers of the cutoff are removed. Use scalar distribution pullback, so that ordinary functions pull back by composition.

For an orientation-preserving analytic coordinate f , put

$$H_f(t) = f(t)/t, \quad \lambda_f(t) = \log H_f(t), \quad C_f[h] = f^*(\text{FP}_x h) - \text{FP}_t(h \circ f).$$

The residue law proved in the incoming reports is

$$\langle C_f[h], \phi \rangle = \sum_{q,n} \frac{c_{q,n}}{n+1} \text{Res}_{t=0} f(t)^{-q} \lambda_f(t)^{n+1} \phi(t) dt. \quad (4.2)$$

It is enough to keep the finite Taylor jet of the smooth test function inside the residue. Integrable terms create no discrepancy. Equivalently, for a single monomial,

$$C_f[x^{-q} \log^n x] = n! \sum_{j=0}^{q-1} \frac{(-1)^j}{j!} [z^{n+1} t^{q-1-j}] H_f(t)^{z-q} \delta_0^{(j)}. \quad (4.3)$$

This is the same coefficient law, because $[z^{n+1}] H_f^{z-q} = H_f^{-q} \lambda_f^{n+1} / (n+1)!$.

For completeness, the analytic justification is short. Continue x_+^{z-q} after subtracting the test-function Taylor polynomial through degree $q-1$. Only the term of degree $q-1$ has denominator z near zero. After pullback, its numerator depends on z through H_f^{z-q} . The constant term of the cutoff coefficient omits the quantity $n![z^{n+1}]$ of that numerator, whereas the meromorphic coefficient retains it. Every other Taylor denominator is nonzero at $z=0$; the remainder is normally integrable there. This proves (4.3), including its sign and factorials.

4.2 The complete classification

For an integer $d \geq 1$, write

$$\mathcal{G}_d = \{f : f(0) = 0, \quad f(t) = t + O(t^{d+1})\}.$$

These are germs with identity Taylor jet through degree d .

Theorem 4.1 (Classification by pole order and logarithmic degree). *For the germ (4.1) and an integer $d \geq 1$, the following conditions are equivalent.*

1. $C_f[h] = 0$ for every $f \in \mathcal{G}_d$.
2. $C_{f_a}[h] = 0$ for every sufficiently small real a , where $f_a(t) = t(1 + at^d)$.
3. $c_{q,n} = 0$ whenever $q > d(n+1)$.

Equivalently, if $h = \sum_{n=0}^L a_n(x) \log^n x$, the pole order of a_n is at most $d(n+1)$ for every n . For $d=1$, the exact condition is $\text{poleord}(a_n) \leq n+1$.

Proof. The first condition implies the second. To prove that the third implies the first, note that $\lambda_f(t) = O(t^d)$ for $f \in \mathcal{G}_d$. The q, n summand of the residue integrand in (4.2) has order at least $d(n+1) - q \geq 0$. It is analytic at zero, so its residue against any smooth test function is zero.

It remains to prove necessity; this is not a consequence of estimating the order of one monomial alone. For $f_a = t(1 + at^d)$, (4.3) becomes

$$C_{f_a}[x^{-q} \log^n x] = n! \sum_{k=1}^{\lfloor (q-1)/d \rfloor} \frac{(-1)^{q-1-dk} a^k}{(q-1-dk)!} [z^{n+1}] \binom{z-q}{k} \delta_0^{(q-1-dk)}. \quad (4.4)$$

The term at $k=0$ vanishes because $n+1 \geq 1$. For a fixed power a^k and derivative $\delta_0^{(j)}$, the pole order is uniquely determined by $q = dk + j + 1$. Therefore different pole orders cannot cancel this coefficient.

Fix q and let $K = \lfloor (q-1)/d \rfloor$. The coefficient of $a^k \delta_0^{(q-1-dk)}$ vanishes for $k = 1, \dots, K$ only if

$$\sum_{n=0}^{k-1} c_{q,n} n! [z^{n+1}] \binom{z-q}{k} = 0, \quad k = 1, \dots, K. \quad (4.5)$$

This is triangular in $c_{q,0}, \dots, c_{q,K-1}$, and its diagonal coefficient at $c_{q,k-1}$ is

$$(k-1)! [z^k] \binom{z-q}{k} = \frac{1}{k} \neq 0.$$

Induction on k gives $c_{q,0} = \dots = c_{q,K-1} = 0$. Those indices are exactly the pairs satisfying $q > d(n+1)$. This proves the third condition and completes the equivalence. $\square$

A useful consequence is that the least required tangency order can be read directly from the singular expansion:

$$d_{\min}(h) = \max_{c_{q,n} \neq 0} \left\lfloor \frac{q}{n+1} \right\rfloor. \quad (4.6)$$

For a germ with no singular term, every coordinate is harmless. For a nonzero singular germ, (4.6) is the least positive integer d such that matching the identity jet through degree d removes every coordinate defect.

Corollary 4.2 (Exact dimension of the obstruction space). *Within pole order at most R and logarithmic degree at most L , the codimension of the invariant subspace is*

$$\sum_{q=1}^R \min \left(L+1, \left\lfloor \frac{q-1}{d} \right\rfloor \right). \quad (4.7)$$

The one-parameter family in (4.4), with its polynomial dependence on a retained, detects all these obstructions.

Proof. Theorem 4.1 sets precisely the listed independent coordinates $c_{q,n}$ to zero. Equation (4.5) proves that their obstruction functionals are independent. $\square$

The reference to the whole parameter family matters: a single numerical choice of a can combine coefficients belonging to different pole orders. The theorem uses an exact polynomial identity in a .

4.3 Sharp Stieltjes thresholds for any order of tangency

Use the usual convention $\gamma_n(x) = x^{-1} \log^n x + \gamma_n(1+x)$. Let $T_{n,r} = \text{FP}_x \gamma_n^{(r)}(x)$ be the finite part of the pointwise argument derivative. This object is distinguished from $D^r(\text{FP}_x \gamma_n)$, whose harmonic contact term was computed in the incoming reports.

Theorem 4.3 (Exact threshold for Stieltjes argument derivatives). *For $r, n \geq 0$ and $d \geq 1$,*

$$f^* T_{n,r} = \text{FP}_t(\gamma_n^{(r)} \circ f) \quad \text{for every } f \in \mathcal{G}_d$$

holds if and only if

$$d \geq r+1 \quad \text{or} \quad n \geq r + \left\lfloor \frac{r}{d} \right\rfloor. \quad (4.8)$$

When $1 \leq d \leq r$, set $K = \lfloor r/d \rfloor$, $e = r - dK$, and $n_ = r + K - 1$. The first excluded index has the exact defect*

$$C_{f_a}[\gamma_{n_*}^{(r)}] = \frac{(-1)^e n_*!}{e! K!} a^K \delta_0^{(e)}, \quad f_a(t) = t(1 + at^d). \quad (4.9)$$

Thus every threshold in (4.8) is attained.

Proof. The singular part is

$$\gamma_n^{(r)}(x) = (-1)^r r! x^{-r-1} n! [z^n] \left\{ \prod_{j=1}^r \left(1 - \frac{z}{j}\right) e^{z \log x} \right\} + \gamma_n^{(r)}(1+x). \quad (4.10)$$

Every elementary symmetric coefficient of $1, 1/2, \dots, 1/r$ is nonzero. Hence its smallest logarithmic degree is $\max(n-r, 0)$, and the coefficient at that degree is nonzero. The classification theorem gives the exact condition

$$r+1 \leq d(\max(n-r, 0) + 1).$$

If $d \geq r+1$ this holds for all n . Otherwise it requires $n \geq r$ and is equivalent to the second inequality in (4.8).

At $n = n_*$, the smallest logarithmic degree is $K-1$, with coefficient $n_*/(K-1)!$ in (4.10). Every higher logarithmic degree is invariant. In (4.4), only $k = K$ survives for this smallest degree, and its derivative index is e . Its coefficient is $(-1)^e a^K/(e!K)$. Multiplication by $n_*/(K-1)!$ proves (4.9). $\square$

For $d = 1$ this recovers the incoming sharp threshold $n \geq 2r$ and its coefficient $(2r-1)!a^r \delta_0/r!$. It additionally gives, for example,

$$C_{t+at^3}[\gamma_3^{(3)}] = -6a\delta'_0, \quad C_{t+at^3}[\gamma_5^{(4)}] = 60a^2\delta_0,$$

while $\gamma_4^{(3)}$ and $\gamma_6^{(4)}$ have no defect under any coordinate $f(t) = t + O(t^3)$.

4.4 Consequences for coincident Stieltjes products

The invariant space is linear but is generally not closed under pointwise multiplication: x^{-1} is invariant under every unit-tangent coordinate, whereas x^{-2} is not. The classification nevertheless provides a useful exact product criterion.

Corollary 4.4 (Products with at most one digamma factor). *Let $m_1, \dots, m_s$ be nonnegative integers, at most one of which is zero. Then the specified pointwise-product finite part satisfies*

$$f^* \text{FP}_x \left(\prod_{j=1}^s \gamma_{m_j}(x) \right) = \text{FP}_t \left(\prod_{j=1}^s \gamma_{m_j}(f(t)) \right) \quad (f \in \mathcal{G}_1). \quad (4.11)$$

In particular, all powers $\text{FP}_x \gamma_m(x)^s$ with $m \geq 1$ have this invariance.

Proof. Choose the singular term from a nonempty subset S of the factors. Before expanding the remaining analytic factors, the resulting term is

$$x^{-|S|} \log^{\sum_{j \in S} m_j} x \prod_{j \notin S} \gamma_{m_j}(1+x).$$

Its logarithmic degree is at least $|S| - 1$. Taylor expansion of the analytic factor can only decrease the pole order. Thus every resulting singular monomial satisfies $q \leq n+1$, which is exactly the criterion of Theorem 4.1 for $d = 1$. $\square$

The restriction on the number of zero indices cannot be replaced by a large total Stieltjes index. If $f(t) = t + at^2 + O(t^3)$ and $\gamma_m = \gamma_m(1)$, direct use of the same classification gives

$$C_f[\gamma_0^2] = a\delta_0, \quad (4.12)$$

$$C_f[\gamma_0^2 \gamma_1] = \left(a\gamma_1 + \frac{a^2}{2} \right) \delta_0, \quad (4.13)$$

$$C_f[\gamma_0^2 \gamma_m] = a\gamma_m \delta_0 \quad (m \geq 2). \quad (4.14)$$

Indeed, the mixed singular–singular–regular term in the last two products is $\gamma_m x^{-2}$. The fully singular term is $x^{-3} \log^m x$: it contributes $a^2 \delta_0 / 2$ when $m = 1$ and nothing when $m \geq 2$. All other terms meet the invariant criterion. These are identities for a specified finite part of the ordinary product; they do not postulate a product of distributions.

Verification. The accompanying exact script compares the residue construction with (4.4), computes the rank of the finite obstruction map, and checks the Stieltjes criterion and its sharp coefficient through several pole and tangency orders. These are independent algebraic audits of signs and finite formulas. The all-order conclusions follow from the proofs above.

5 Exact normal spectral jets at integer inner orders

Write

$$\mathcal{T}(A, B, C; X, Y) = \sum_{m, n \geq 1} \frac{X^m Y^n}{m^A n^B (m+n)^C}.$$

The orders are independent complex variables, and powers of positive integers use real logarithms. The question in the incoming *Resonance and Coordinate Transport* report asks for finite reductions of the normal C -derivatives at $C = 0$, beginning with $A, B \in \{0, 1\}$. The negative- C value identity by itself does not answer that question. The reduction below keeps C variable from the outset.

The underlying positive-integer partial-fraction identity is classical: Zhao explicitly uses it to reduce colored Tornheim values to double polylogarithms [13]. Our use of that identity retains the complete complex C -dependence, specifies the continuation required by the normal jets, and combines it with a rational-function theorem covering arbitrary depth, integer weights, and coincident colors. These are exact reductions to stated function classes. No arithmetic independence or global priority claim is inferred from them.

For the double polylogarithm use the convention

$$\mathrm{Li}_{u,v}(z, w) = \sum_{k > n \geq 1} \frac{z^k w^n}{k^u n^v}. \quad (5.1)$$

Initially the colors can be taken with $|X|, |Y| < 1$ and $X, Y \neq 0$, so that the reindexed series below converge locally normally for all orders. On $|X| = |Y| = 1$ we assume $X, Y \neq 1$ and use radial analytic continuation: replace (X, Y) by $(\rho X, \rho Y)$ and let $\rho \uparrow 1$. In (5.1) this means damping its outer color by ρ . This convention is unambiguous for the combinations appearing here. Indeed

$$\mathcal{T}(0, B, C; X, Y) = \frac{1}{\Gamma(C)} \int_0^\infty t^{C-1} \mathrm{Li}_0(X e^{-t}) \mathrm{Li}_B(Y e^{-t}) dt$$

for $\Re C > 0$. The product is analytic at $t = 0$ and exponentially decreasing at infinity. Taylor subtraction at zero, followed by the zeros of $1/\Gamma(C)$, gives an entire continuation in (B, C) . The same argument applies with arbitrary fixed integer first orders, and identifies the radial continuation locally uniformly in the orders. Thus all the order derivatives used below exist.

Theorem 5.1 (Integer inner orders and all normal jets). *For positive integers A, B , one has the identity of entire functions of C at nontrivial unit colors*

$$\begin{aligned} \mathcal{T}(A, B, C; X, Y) &= \sum_{j=1}^A \binom{A+B-j-1}{B-1} \mathrm{Li}_{C+A+B-j,j}(Y, X/Y) \\ &\quad + \sum_{j=1}^B \binom{A+B-j-1}{A-1} \mathrm{Li}_{C+A+B-j,j}(X, Y/X). \end{aligned} \quad (5.2)$$

For $M \geq 0$ an integer and arbitrary $B \in \mathbb{C}$,

$$\mathcal{T}(-M, B, C; X, Y) = \sum_{j=0}^M (-1)^j \binom{M}{j} \text{Li}_{C-M+j, B-j}(X, Y/X). \quad (5.3)$$

Every fixed C -derivative can be applied to these formulas. In (5.3), every fixed B -derivative is also valid. Together with exchange of the two inner variables, these formulas give a finite reduction for every pair of integer inner orders.

Proof. For positive m, n ,

$$\frac{1}{m^A n^B} = \sum_{j=1}^A \binom{A+B-j-1}{B-1} \frac{1}{m^j (m+n)^{A+B-j}} + \sum_{j=1}^B \binom{A+B-j-1}{A-1} \frac{1}{n^j (m+n)^{A+B-j}}.$$

It follows by repeated use of $1/(mn) = (1/m + 1/n)/(m+n)$, or by induction on $A+B$. Multiply by the colors and $(m+n)^{-C}$ and put $k = m+n$. The first sum has color $Y^k(X/Y)^m$, and the second has $X^k(Y/X)^n$. This proves (5.2) in an absolutely convergent region. For (5.3), expand $(k-n)^M = \sum_j (-1)^j \binom{M}{j} k^{M-j} n^j$ after the same reindexing. Local normal convergence inside the bidisc and then the specified analytic continuation prove both identities and the derivative claims. $\square$

Here is the promised complete $A, B \in \{0, 1\}$ table. A superscript $[r]$ on a double polylogarithm in this table differentiates only its first order. For $X \neq Y$,

$$\partial_C^r \mathcal{T}(0, 0, C; X, Y) = \frac{Y \text{Li}_C^{[r]}(X) - X \text{Li}_C^{[r]}(Y)}{X - Y}, \quad (5.4)$$

$$\partial_C^r \mathcal{T}(0, 1, C; X, Y) = \text{Li}_{C,1}^{[r]}(X, Y/X), \quad (5.5)$$

$$\partial_C^r \mathcal{T}(1, 0, C; X, Y) = \text{Li}_{C,1}^{[r]}(Y, X/Y), \quad (5.6)$$

$$\partial_C^r \mathcal{T}(1, 1, C; X, Y) = \text{Li}_{C+1,1}^{[r]}(X, Y/X) + \text{Li}_{C+1,1}^{[r]}(Y, X/Y). \quad (5.7)$$

The confluent version of (5.4) is

$$\partial_C^r \mathcal{T}(0, 0, C; z, z) = \text{Li}_{C-1}^{[r]}(z) - \text{Li}_C^{[r]}(z). \quad (5.8)$$

All $C = -N$ and $C = 0$ substitutions are now legitimate. Formula (5.7) is a finite depth-two answer; it does not claim a reduction of the remaining depth-two order derivatives to ordinary polylogarithms.

Remark 5.2 (The permitted derivative directions). In (5.2), the first two orders are fixed integers. Its finite expression does not authorize differentiating with respect to A or B . In particular, it does not reduce the full mixed jet $\partial_A \partial_B \partial_C \mathcal{T}$ by formal differentiation of the finite sums. In (5.3), B really is free, so B -derivatives are allowed, while A remains the fixed integer $-M$. This is the same distinction that prevents differentiating a negative- C value formula transversely without further information.

5.1 Rational generating functions give depth-one closure

Let $d \geq 1$, let p_j be positive integers, let a_j be nonnegative integers, and let X_j be nonzero complex numbers with $|X_j| \leq 1$. Define, initially for $\Re C > a_1 + \dots + a_d + d$,

$$Z(C) = \sum_{m_1, \dots, m_d \geq 1} \frac{\prod_{j=1}^d m_j^{a_j} X_j^{m_j}}{(p_1 m_1 + \dots + p_d m_d)^C}. \quad (5.9)$$

Consider its coefficient generating function

$$R(t) = \prod_{j=1}^d \text{Li}_{-a_j}(X_j t^{p_j}) = \sum_{n \geq 1} r_n t^n. \quad (5.10)$$

The elementary identity $\text{Li}_{-a}(z) = (z d/dz)^a \{z/(1-z)\}$ proves that R is rational and bounded at infinity. Write its partial fractions in the form

$$R(t) = c_\infty + \sum_{\alpha} \sum_{h=1}^{M_{\alpha}} \frac{c_{\alpha,h}}{(1-\alpha t)^h}, \quad (5.11)$$

where each α is a root of some equation $\alpha^{p_j} = X_j$. A permissible multiplicity bound is

$$M_{\alpha} = \sum_{j: \alpha^{p_j} = X_j} (a_j + 1).$$

This is an upper bound: zeros of other factors can cancel some poles. In that event the corresponding highest coefficients are zero. There is no positive-degree polynomial part because R is bounded at infinity. An exact coefficient formula, also valid at repeated roots, is

$$c_{\alpha,h} = [u^{M_{\alpha}-h}] u^{M_{\alpha}} R((1-u)/\alpha). \quad (5.12)$$

The function whose coefficient is extracted is holomorphic at $u = 0$.

Theorem 5.3 (Arbitrary depth, weights, multiplicities, and normal jets). *Set*

$$e_{h,r} = [u^r] \prod_{\nu=1}^{h-1} (1 + u/\nu), \quad (5.13)$$

$$d_{\alpha,r} = \sum_{h=r+1}^{M_{\alpha}} c_{\alpha,h} e_{h,r}. \quad (5.14)$$

Empty products equal one. Then

$$Z(C) = \sum_{\alpha} \sum_{r=0}^{M_{\alpha}-1} d_{\alpha,r} \text{Li}_{C-r}(\alpha). \quad (5.15)$$

This is the meromorphic continuation of (5.9), and every C -derivative is obtained by replacing each polylogarithm by the corresponding order derivative.

If no X_j equals one, the continuation is entire. In general its only possible poles are simple poles at $C = r + 1$ from the $\alpha = 1$ sector, with residue $d_{1,r}$. For $k \geq 0$, the constant Laurent coefficient of $Z^{(k)}$ at an integer $m \geq 1$ is

$$\text{CT}_{C=m} Z^{(k)}(C) = (-1)^k d_{1,m-1} \gamma_k + \sum_{(\alpha,r) \neq (1,m-1)} d_{\alpha,r} \text{Li}_{m-r}^{[k]}(\alpha). \quad (5.16)$$

Missing coefficients are zero; for $\alpha = 1$ the nonsingular terms mean ordinary Riemann-zeta derivatives.

Proof. The original multiple sum converges absolutely in the stated half-plane. Grouping it according to $n = \sum p_j m_j$ gives $Z(C) = \sum_{n \geq 1} r_n n^{-C}$. The binomial series gives

$$[t^n](1-\alpha t)^{-h} = \alpha^n \binom{n+h-1}{h-1} = \alpha^n \prod_{\nu=1}^{h-1} (1+n/\nu) = \alpha^n \sum_{r=0}^{h-1} e_{h,r} n^r.$$

For $n \geq 1$ the constant c_∞ contributes nothing. Substitution proves (5.15) where the series converges absolutely and hence by continuation. If $\alpha \neq 1$ and $|\alpha| \leq 1$, the fixed-color polylogarithm is entire in its order; for $\alpha = 1$ it is ζ . The pole and derivative statements follow. Finally

$$\zeta(1+v) = v^{-1} + \sum_{j \geq 0} \frac{(-1)^j \gamma_j}{j!} v^j$$

shows that the constant coefficient after k differentiations is $(-1)^k \gamma_k$. The other terms are holomorphic at the specified integer, proving (5.16). $\square$

The harmonic-number coefficients in this theorem can be computed by the finite-degree expansion

$$\prod_{\nu=1}^{h-1} (1 + u/\nu) = \exp \left(\sum_{j \geq 1} \frac{(-1)^{j+1}}{j} H_{h-1}^{(j)} u^j \right). \quad (5.17)$$

For example $e_{h,0} = 1$, $e_{h,1} = H_{h-1}$, and $e_{h,2} = (H_{h-1}^2 - H_{h-1}^{(2)})/2$. All multiplicity corrections are consequently finite polynomials in generalized harmonic numbers. Coincidence in this statement concerns the poles of the rational coefficient generating function. It does not identify two independently specified spatial finite parts.

5.2 Roots of unity, Gamma values, and an explicit weighted example

For a nontrivial q th root of unity α , residue-class summation gives

$$\text{Li}_s(\alpha) = q^{-s} \sum_{j=1}^q \alpha^j \zeta(s, j/q). \quad (5.18)$$

It follows first where the series converge absolutely and then by continuation. In particular, for $s \neq 1$,

$$\text{Li}_s^{[k]}(\alpha) = q^{-s} \sum_{j=1}^q \alpha^j \sum_{\ell=0}^k \binom{k}{\ell} (-\log q)^{k-\ell} \zeta^{[\ell]}(s, j/q). \quad (5.19)$$

At $s = 1$, cancellation of the common pole gives instead

$$\text{Li}_1^{[k]}(\alpha) = \frac{1}{q} \sum_{j=1}^q \alpha^j \sum_{\ell=0}^k \binom{k}{\ell} (-\log q)^{k-\ell} (-1)^\ell \gamma_\ell(j/q). \quad (5.20)$$

The cancellation is performed before evaluation. Applying Lerch's formula $\zeta'(0, a) = \log \Gamma(a) - \frac{1}{2} \log(2\pi)$ [10, Eq. 25.11.18] gives

$$\text{Li}_0^{[1]}(\alpha) = \sum_{j=1}^q \alpha^j \log \Gamma(j/q) - \log q \frac{\alpha}{1-\alpha}. \quad (5.21)$$

Thus simple poles in (5.11) yield a finite log-Gamma evaluation for $Z'(0)$ at roots of unity. Repeated poles additionally introduce the explicitly listed Hurwitz derivatives at negative integers from (5.19); they should not be silently discarded.

For a concrete unequal-weight example, put

$$Z_{2,3}(C) = \sum_{m,n \geq 1} \frac{(-1)^{m+n}}{(2m+3n)^C} \quad (\Re C > 2).$$

Its rational generating function has the exact certificate

$$\frac{t^5}{(1+t^2)(1+t^3)} = \frac{t^5 - t^7 - t^8 + t^9 + t^{10} - t^{12}}{1 - t^{12}}. \quad (5.22)$$

Consequently, for every C by entire continuation,

$$Z_{2,3}(C) = 12^{-C} \{ \zeta(C, 5/12) - \zeta(C, 7/12) - \zeta(C, 2/3) \\ + \zeta(C, 3/4) + \zeta(C, 5/6) - \zeta(C, 1) \}. \quad (5.23)$$

The six residues at $C = 1$ cancel. In particular,

$$Z'_{2,3}(0) = \log \frac{\Gamma(5/12)\Gamma(3/4)\Gamma(5/6)}{\Gamma(7/12)\Gamma(2/3)} - \frac{1}{4} \log 12. \quad (5.24)$$

An entirely convergent integral version, with no formal double sum at $C = 0$, is

$$\int_0^1 \left\{ \frac{1}{(e^{2t} + 1)(e^{3t} + 1)} - \frac{1}{4} \right\} \frac{dt}{t} + \int_1^\infty \frac{dt}{t(e^{2t} + 1)(e^{3t} + 1)} + \frac{\gamma}{4} \\ = \log \frac{\Gamma(5/12)\Gamma(3/4)\Gamma(5/6)}{\Gamma(7/12)\Gamma(2/3)} - \frac{1}{4} \log 12. \quad (5.25)$$

To verify the integral, insert the Mellin representation for $\Re C > 0$, subtract its constant $1/4$ at zero, and use $1/\Gamma(C) = C + \gamma C^2 + O(C^3)$.

For a fully confluent example, d identical colors with unit weights and zero inner orders give

$$\sum_{m_1, \dots, m_d \geq 1} \frac{z^{m_1 + \dots + m_d}}{(m_1 + \dots + m_d)^C} = \frac{1}{(d-1)!} \sum_{r=0}^{d-1} s(d, r+1) \text{Li}_{C-r}(z), \quad (5.26)$$

where $s(d, j)$ are the signed Stirling numbers of the first kind. Indeed the coefficient at total index n counts compositions and equals $\binom{n-1}{d-1}$, including its zero values for $1 \leq n < d$. This proves the identity and all its normal derivatives. For $d = 3$, $z = -1$, and differentiation at $C = 0$, it yields

$$\frac{d}{dC} \sum_{m, n, k \geq 1} \frac{(-1)^{m+n+k}}{(m+n+k)^C} \Big|_{C=0}^{\text{cont}} = -\frac{7\zeta(3)}{8\pi^2} - \frac{9}{2}\zeta'(-1) - \frac{1}{2} \log \pi. \quad (5.27)$$

Here “cont” explicitly means continuation from $\Re C > 3$. The reduction uses $\text{Li}_s(-1) = (2^{1-s} - 1)\zeta(s)$, $\zeta'(0) = -\frac{1}{2} \log(2\pi)$, and $\zeta'(-2) = -\zeta(3)/(4\pi^2)$.

5.3 An exact primitive hierarchy

Keep the coefficients r_n of (5.10) and put $Z(C; z) = \sum_{n \geq 1} r_n z^n / n^C$ for $|z| < 1$. The proof of Theorem 5.3 gives

$$Z(C; z) = \sum_{\alpha, r} d_{\alpha, r} \text{Li}_{C-r}(\alpha z).$$

For every nonnegative integer k ,

$$\int_0^z Z^{(k)}(C; t) \frac{dt}{t} = Z^{(k)}(C+1; z), \quad z \frac{d}{dz} Z^{(k)}(C; z) = Z^{(k)}(C-1; z). \quad (5.28)$$

The integrand is holomorphic at zero after division by t , and the primitive is fixed by its zero value there. These identities follow termwise in the unit disc and continue along any common path avoiding the polylogarithmic singularities. This gives exact antiderivatives of every normal jet without changing its spectral direction.

5.4 Independent algebraic and numerical checks

The accompanying script `verify_spectral.py` performs 402 exact finite checks: 25 positive-order partial fractions, 12 composition polynomials, 105 harmonic–Stirling coefficients, seven reconstructions of rational generating functions, 252 independently convolved coefficients, and the cyclotomic certificate (5.22). The rational cases include unequal weights, repeated roots, multiplicity six, and cancellation of an otherwise apparent pole. Nine independently evaluated convergent Mellin integrals at 60 decimal digits check the first normal jets for distinct colors i , $-i$, repeated color -1 in depths two and three, the unequal weights $(2, 3)$, and the root-Gamma formula for roots of orders 3, 4, 5, 6, 12. The largest observed discrepancy is 4.7×10^{-61} . The exact algebraic checks and analytic proofs are separate from these floating-point diagnostics; the latter are not interval-certified error bounds. Inputs, formulas, and full outputs are supplied in `spectral_results.json`.

6 The first two jets on every admissible ray through the origin

The uncolored Tornheim function is not jointly holomorphic at the origin. Consequently $\mathcal{T}(0, 0, 0; 1, 1)$ is not a value that can be assigned independently of a restriction convention. Borwein and Dilcher established this explicitly for the family $\mathcal{T}(t, t, \tau t; 1, 1)$ and evaluated its first derivative [9, Theorems 2.7 and 3.1]. The diagonal value $1/3$ is also recorded by Romik [14, Theorem 1.3], who credits the simple diagonal derivative to Borwein and Dilcher.

We give the asymmetric first jet and an exact second-jet formula for every ray avoiding the two polar directions. Let

$$\mathcal{T}(A, B, C) = \mathcal{T}(A, B, C; 1, 1), \quad L = \log(2\pi), \quad W_{a,b,c}(t) = \mathcal{T}(at, bt, ct).$$

Each restriction is formed away from the polar set and then continued as a one-variable function. It is not the evaluation of a nonexistent joint Taylor series at $(0, 0, 0)$.

Theorem 6.1 (Asymmetric ray values and quadratic jets). *Let $a, b, c \in \mathbb{C}$ satisfy $(a+c)(b+c) \neq 0$, and define*

$$\mathcal{B}(a, b, c) = \frac{c(c^2 + ab - a^2 - b^2)}{(a+c)(b+c)}. \quad (6.1)$$

Then $W_{a,b,c}$ has a removable singularity at $t = 0$, and

$$W_{a,b,c}(0) = \frac{1}{4} + \frac{c}{12(a+c)} + \frac{c}{12(b+c)}, \quad (6.2)$$

$$W'_{a,b,c}(0) = \frac{a+b+2c}{4}L + \mathcal{B}(a, b, c)\zeta'(-1), \quad (6.3)$$

$$W''_{a,b,c}(0) = \frac{2ab+c(a+b)}{4}L^2 - \frac{a^2+b^2+2c(a+b)+2c^2}{2}\zeta''(0) + (a+b+c)\mathcal{B}(a, b, c)\zeta''(-1). \quad (6.4)$$

The equal-inner-order specialization is especially compact:

$$\begin{aligned} W_{1,1,\tau}(0) &= \frac{5\tau+3}{12(\tau+1)}, \\ W'_{1,1,\tau}(0) &= \frac{\tau+1}{2}L + \frac{\tau(\tau-1)}{\tau+1}\zeta'(-1), \\ W''_{1,1,\tau}(0) &= \frac{\tau+1}{2}L^2 - (\tau+1)^2\zeta''(0) + \frac{\tau(\tau-1)(\tau+2)}{\tau+1}\zeta''(-1). \end{aligned} \quad (6.5)$$

The first two lines recover the cited classical result; the third is the quadratic extension proved here. In particular,

$$\left. \frac{d^2}{dt^2} \mathcal{T}(t, t, t) \right|_{t=0} = \log^2(2\pi) - 4\zeta''(0). \quad (6.6)$$

6.1 A local meromorphic decomposition

Put $F(A, B; x) = \text{Li}_A(e^{-x}) \text{Li}_B(e^{-x})$. Jonquière's convergent expansion for $0 < x < 2\pi$ gives

$$\text{Li}_A(e^{-x}) = \Gamma(1-A)x^{A-1} + \sum_{j \geq 0} \zeta(A-j) \frac{(-x)^j}{j!}$$

for A near zero; all powers of x use its real logarithm. Subtract the following six terms:

$$\begin{aligned} S(A, B; x) = & \Gamma(1-A)\Gamma(1-B)x^{A+B-2} + \Gamma(1-A)\zeta(B)x^{A-1} \\ & + \Gamma(1-B)\zeta(A)x^{B-1} - \Gamma(1-A)\zeta(B-1)x^A \\ & - \Gamma(1-B)\zeta(A-1)x^B + \zeta(A)\zeta(B). \end{aligned} \quad (6.7)$$

On a sufficiently small parameter polydisc the remainder and each fixed parameter derivative are $O(x^{1-\delta}(1+|\log x|^N))$ for some $\delta < 1$. Consequently the function

$$\begin{aligned} H(A, B, C) = & \int_0^1 x^{C-1} \{F(A, B; x) - S(A, B; x)\} dx + \int_1^\infty x^{C-1} F(A, B; x) dx \\ & + \frac{\Gamma(1-A)\Gamma(1-B)}{A+B+C-2} + \frac{\Gamma(1-A)\zeta(B)}{A+C-1} + \frac{\Gamma(1-B)\zeta(A)}{B+C-1} \end{aligned} \quad (6.8)$$

is jointly holomorphic near $(0, 0, 0)$ and symmetric in A, B . The Mellin formula now gives the exact meromorphic decomposition

$$\begin{aligned} \mathcal{T}(A, B, C) = & \frac{1}{\Gamma(1+C)} \left\{ \zeta(A)\zeta(B) - \frac{C\Gamma(1-A)\zeta(B-1)}{A+C} \right. \\ & \left. - \frac{C\Gamma(1-B)\zeta(A-1)}{B+C} + CH(A, B, C) \right\}. \end{aligned} \quad (6.9)$$

To justify it, integrate each term of (6.7) explicitly in a common Mellin convergence region, and then continue; the subtracted integrals in (6.8) supply the required local continuation. The only polar hyperplanes meeting the origin are now displayed. On an admissible ray, their quotients are constants, so the apparent one-variable singularity is removable.

6.2 The diagonal input

The diagonal identity (6.6) has already been proved in Theorem 2.3 by comparing the local cubic residue with its Hurwitz Fourier representation. That comparison uses only the coefficient of the simple pole: the unknown cubic finite part first enters one order later. There is therefore no dependence on a numerical value of the cubic integral in the argument below.

6.3 Completing the asymmetric calculation

The exact restriction

$$\mathcal{T}(0, 0, C) = \zeta(C-1) - \zeta(C) \quad (6.10)$$

follows by counting the $k-1$ positive pairs summing to k in its absolute-convergence half-plane. Restricting (6.9) to this axis gives

$$\Gamma(1+C)\{\zeta(C-1) - \zeta(C)\} = \frac{5}{12} + CH(0, 0, C).$$

Hence, writing all partial derivatives of H below at the origin,

$$H(0, 0, 0) = \zeta'(-1) + \frac{L}{2} - \frac{5\gamma}{12}, \quad (6.11)$$

$$H_C = \frac{\zeta''(-1) - \zeta''(0)}{2} - \gamma \left(\zeta'(-1) + \frac{L}{2} \right) + \frac{5}{24} \{\gamma^2 + \zeta(2)\}. \quad (6.12)$$

The symmetry $H_A = H_B$, together with (6.6) in (6.9), then gives

$$H_A = H_B = \frac{L^2}{8} - \frac{\zeta''(0)}{2} + \gamma\zeta'(-1) - \frac{\gamma L}{4} - \frac{\gamma^2 + \zeta(2)}{24}. \quad (6.13)$$

These equations determine the entire constant and linear Taylor polynomial of the jointly holomorphic remainder.

To prove Theorem 6.1, substitute $A = at, B = bt, C = ct$ in (6.9), use (6.11)–(6.13), and expand to second order. The constant is (6.2). The coefficient of $\zeta'(-1)$ in the first derivative is

$$c - \frac{bc}{a+c} - \frac{ac}{b+c} = \mathcal{B}(a, b, c),$$

and all Euler-constant terms cancel, proving (6.3). In the second derivative, the surviving coefficients of $L^2, \zeta''(0), \zeta''(-1)$ are respectively

$$\frac{2ab + c(a+b)}{4}, \quad -\frac{a^2 + b^2 + 2c(a+b) + 2c^2}{2}, \quad (a+b+c)\mathcal{B}(a, b, c).$$

The coefficients of $\gamma^2, \zeta(2), \gamma\zeta'(-1)$ and γL cancel identically. This proves (6.4) and completes the theorem.

For example, the diagonal and the third-order-parameter axis give

$$W_{1,1,1}(0) = \frac{1}{3}, \quad W_{0,0,1}(0) = \frac{5}{12}.$$

For the asymmetric ray $(a, b, c) = (1, 2, 3)$ one obtains

$$\begin{aligned} W_{1,2,3}(0) &= \frac{29}{80}, \\ W'_{1,2,3}(0) &= \frac{9}{4}L + \frac{9}{10}\zeta'(-1), \\ W''_{1,2,3}(0) &= \frac{13}{4}L^2 - \frac{41}{2}\zeta''(0) + \frac{27}{5}\zeta''(-1). \end{aligned}$$

These are explicit directional identities, and also a concrete audit test for any routine that substitutes integer orders before differentiating. A routine returning the same value for the first two restrictions has erased the specified limiting direction.

Independent numerical check. The accompanying `verify_rays.py` integrates the convergent Jonquière series on $(0, 1)$ and an incomplete-Gamma double series on $(1, \infty)$. It does not evaluate a cubic finite-part formula. At 65-digit working precision, seven successive symmetric step sizes and polynomial extrapolation reproduce the first two derivatives on the rays $(1, 1, 1)$, $(1, 2, 3)$, $(2, 3, 1)$, and $(1, -2, 3)$. The largest reported absolute discrepancy is below 2.6×10^{-37} . These are diagnostics of the analytic proof, with truncation and floating-point errors not certified by interval arithmetic.

7 Two independent exponents at a harmonic resonance

The fully shifted height-one theorem in the incoming exact-identities report varies the outer exponent and keeps every inner exponent equal to one. Its final research section asks what survives when inner exponents move independently. We answer a precise part of that question: the first inner exponent may vary independently of the outer exponent, with an arbitrary number of subsequent unit exponents. Every principal Laurent coefficient and the finite part along every transverse affine direction have a finite Gamma–Bernoulli formula. A polynomial in the common shift supplies the same correction in all directions.

The meromorphic continuation of multiple zeta functions and the dependence of specializations on the approach direction are classical [15, 16]. In particular, Matsumoto, Onozuka and Wakabayashi give algorithms for Laurent expansions at arbitrary integer points, allowing remainder integrals in general. The result below is a compact closed coefficient formula for this particular common-shift family, uniform in its trailing depth and its two-dimensional direction. We do not claim priority for general continuation or multivariable Laurent expansion. We also do not resolve arbitrary independent perturbations of all trailing exponents.

7.1 The family and its finite residue polynomial

Throughout this section, $\Re a > 0$, and powers of $n + a$ use the principal logarithm. Define

$$e_r(n; a) = [u^r] \prod_{j=0}^{n-1} \left(1 + \frac{u}{j+a}\right), \quad p_n(a, u) = \frac{(a+u)_n}{(a)_n}, \quad (7.1)$$

$$Z_r(s, t; a) = \sum_{n_0 > n_1 > \dots > n_{r+1} \geq 0} \frac{1}{(n_0 + a)^s (n_1 + a)^t \prod_{j=2}^{r+1} (n_j + a)}, \quad r \geq 0. \quad (7.2)$$

The final product is empty at $r = 0$. Thus Z_0 is a double Hurwitz zeta function, and $Z_r(s, 1; a)$ is the height-one function with $r + 1$ inner unit exponents. Its depth generator is

$$\mathcal{Z}_a(s, t, u) = \sum_{n \geq 0} p_n(a, u) (n+a)^{-t} \zeta(s, n+a+1) = \sum_{r \geq 0} Z_r(s, t; a) u^r. \quad (7.3)$$

For example, absolute normal convergence holds when $\Re s > 1$ and $\Re(s+t) > 2 + \Re u$; a small closed disc in u can be chosen in any strict version of these inequalities. The estimate $p_n = O(n^{\Re u})$ and the usual Hurwitz-tail estimate prove this claim and justify coefficient extraction. At each fixed depth, $e_r(n; a) = O((1 + \log n)^r)$ gives the corresponding normal convergence.

Let $B_k(x)$ denote Bernoulli polynomials, with $B_1(0) = -1/2$. Define polynomials $q_j(u)$ through the formal series

$$\sum_{j \geq 0} q_j(u) z^j = \exp \left(\sum_{k \geq 1} \frac{(-1)^{k+1} \{B_{k+1}(u) - B_{k+1}(0)\}}{k(k+1)} z^k \right). \quad (7.4)$$

This is a coefficient definition; no convergence of the infinite asymptotic series is asserted. The classical Gamma-ratio expansion [12] reads

$$\frac{\Gamma(x+u)}{\Gamma(x)} \sim x^u \sum_{j \geq 0} q_j(u) x^{-j}. \quad (7.5)$$

In particular,

$$\begin{aligned} q_0 &= 1, & q_1 &= \frac{u(u-1)}{2}, \\ q_2 &= \frac{u(u-1)(u-2)(3u-1)}{24}, & q_3 &= \frac{u^2(u-1)^2(u-2)(u-3)}{48}. \end{aligned}$$

Every finite truncation has a locally uniform remainder in u . Put

$$C_a(u) = \frac{\Gamma(a)}{\Gamma(a+u)}, \quad (7.6)$$

$$A_0(s) = \frac{1}{s-1}, \quad A_1(s) = -\frac{1}{2}, \quad A_{2k}(s) = \frac{B_{2k}}{(2k)!}(s)_{2k-1} \quad (k \geq 1), \quad A_{2k+1}(s) = 0 \quad (k \geq 1), \quad (7.7)$$

$$P_m(s, u) = \sum_{j=0}^{m+1} q_j(u) A_{m+1-j}(s). \quad (7.8)$$

Thus P_m is a finite rational expression, analytic for s near $-m$. Equivalently,

$$P_m(s, u) = \frac{q_{m+1}(u)}{s-1} - \frac{q_m(u)}{2} + \sum_{k=1}^{\lfloor (m+1)/2 \rfloor} \frac{B_{2k}(s)_{2k-1}}{(2k)!} q_{m+1-2k}(u).$$

The first two examples are

$$P_0(s, u) = \frac{u(u-1)}{2(s-1)} - \frac{1}{2}, \quad (7.9)$$

$$P_1(s, u) = \frac{u(u-1)(u-2)(3u-1)}{24(s-1)} - \frac{u(u-1)}{4} + \frac{s}{12}. \quad (7.10)$$

Proposition 7.1 (One moving divisor before depth extraction). *For every integer $m \geq 0$, there is a jointly holomorphic function $\mathcal{H}_{m,a}$ near $(s, t, u) = (-m, 1, 0)$ such that*

$$\mathcal{Z}_a(s, t, u) = \frac{C_a(u)P_m(s, u)}{s+t+m-1-u} + \mathcal{H}_{m,a}(s, t, u). \quad (7.11)$$

The statement is also locally uniform in a on the right half-plane.

Proof. Write $x = n + a$. Since $\zeta(s, x+1) = \zeta(s, x) - x^{-s}$, the Hurwitz asymptotic expansion [10] gives

$$\zeta(s, x+1) \sim \sum_{k \geq 0} A_k(s) x^{1-s-k}.$$

Multiplying a finite truncation by (7.5) yields

$$p_n(a, u) x^{-t} \zeta(s, x+1) \sim C_a(u) \sum_{\ell \geq 0} \left(\sum_{j+k=\ell} q_j(u) A_k(s) \right) x^{1-s-t+u-\ell}.$$

Subtract sufficiently many terms from the summand in (7.3). The remainder series converges normally in a neighborhood of $(-m, 1, 0)$: after retaining L terms, its summands satisfy

$$O(n^{1-\Re(s+t)+\Re u-L}).$$

Choosing $L > m + 2$ suffices on a small neighborhood. Parameter derivatives introduce only fixed powers of $\log n$ after shrinking that neighborhood.

The sums of the subtracted terms are the corresponding finite linear combination of $\zeta(s + t - 1 - u + \ell, a)$. Exactly one can have a pole in this neighborhood: $\ell = m + 1$. Its residue is $C_a(u)P_m(s, u)$, and subtracting its simple pole leaves a holomorphic function. This proves (7.11). The use of finite asymptotic subtraction, with a normally convergent remainder, is what makes this a meromorphic identity rather than a formal asymptotic assertion. $\square$

7.2 An explicit value of the holomorphic remainder

The next lemma is the specialization mechanism from the incoming fully shifted height-one theorem [4]. We include its proof because the direction-independent correction in the main theorem depends on it.

Lemma 7.2 (A finite Stirling specialization). *Let*

$$F_a(w, u) = \sum_{n \geq 0} \frac{p_n(a, u)}{(n+a)^w}, \quad T_m(a, u) = - \sum_{j=1}^{m+1} \left\{ \begin{matrix} m+1 \\ j \end{matrix} \right\} \frac{(a-1)_j}{u+j}.$$

Here $(a-1)^{\underline{j}}$ is a falling factorial and braces denote Stirling numbers of the second kind. For sufficiently small nonzero u , the meromorphic continuation satisfies

$$F_a(-m, u) = T_m(a, u). \quad (7.12)$$

Consequently, with

$$L_m(a, u) = \frac{T_m(a, u) - \zeta(-m, a)}{u}, \quad (7.13)$$

one has

$$\mathcal{H}_{m,a}(-m, 1, u) = L_m(a, u) + \frac{C_a(u)P_m(-m, u)}{u}. \quad (7.14)$$

The right side of (7.14) is holomorphic at $u = 0$.

Proof. The Mellin kernel of F_a is

$$G_a(v, u) = e^{-av} {}_2F_1(1, a+u; a; e^{-v}).$$

Gauss connection at one gives

$$\begin{aligned} G_a(v, u) &= \frac{\Gamma(a)\Gamma(1+u)}{\Gamma(a+u)} \frac{e^{-v}}{(1-e^{-v})^{1+u}} + R_a(v, u), \\ R_a(v, u) &= -\frac{a-1}{1+u} e^{-av} {}_2F_1(1, a+u; 2+u; 1-e^{-v}). \end{aligned}$$

The second kernel is analytic at $v = 0$ and decays exponentially at infinity, locally in the parameters. Mellin subtraction therefore shows that its quotient by $\Gamma(w)$ is entire in w , and has value $(-1)^m m! [v^m] R_a(v, u)$ at $w = -m$. The first kernel has local exponents $-1-u+j$; for small nonzero u its Mellin transform has no pole at $w = -m$, so division by $\Gamma(w)$ makes its value zero there. This proves that the left side of (7.12) is a polynomial in a of degree at most $m+1$.

To identify the polynomial, take a positive integer $a = A$. Direct index shifting in the original Dirichlet series gives

$$F_A(w, u) = \frac{\Gamma(A)\Gamma(1+u)}{\Gamma(A+u)} \left(F_1(w, u) - \sum_{k=1}^{A-1} \frac{(1+u)_{k-1}}{(k-1)! k^w} \right).$$

At $w = -m$, the first term vanishes. The finite identity

$$k^m = \sum_{j=1}^{m+1} \left\{ \begin{matrix} m+1 \\ j \end{matrix} \right\} (k-1)^{\underline{j-1}}$$

and the binomial summation

$$\frac{\Gamma(A)\Gamma(1+u)}{\Gamma(A+u)} \sum_{k=1}^{A-1} \frac{(1+u)_{k-1}}{(k-1)!} (k-1)^{\underline{j-1}} = \frac{(A-1)^{\underline{j}}}{u+j}$$

give $T_m(A, u)$. Both sides are polynomials in a of degree at most $m+1$, and agreement at all positive integers proves (7.12) for every $\Re a > 0$.

Finally, $p_{n+1} - p_n = up_n/(n+a)$ implies, first by absolutely convergent summation,

$$\mathcal{Z}_a(s, 1, u) = \frac{F_a(s, u) - \zeta(s, a)}{u}. \quad (7.15)$$

Meromorphic continuation, (7.11), and (7.12) now give (7.14). Its holomorphy also follows directly from Proposition 7.1. $\square$

The order of operations in (7.12) matters. It restricts w to $-m$ for nonzero u and then continues the resulting function in u . At a crossing of a moving pole it does not give the Laurent finite part obtained by first extracting a coefficient in u . The main theorem keeps these operations in the required order.

For later use put

$$\Delta_{m,d}(a) = (-1)^{d+1} \sum_{j=1}^{m+1} \begin{Bmatrix} m+1 \\ j \end{Bmatrix} \frac{(a-1)^j}{j^{d+1}}. \quad (7.16)$$

Thus $[u^r]L_m(a, u) = \Delta_{m,r+1}(a)$ for $r \geq 0$.

7.3 The complete two-direction finite-part theorem

Theorem 7.3 (All depths and every transverse affine direction). *Let $m, r \geq 0$, $\Re a > 0$, and $\alpha, \beta \in \mathbb{C}$ with $q = \alpha + \beta \neq 0$. Set $\theta = \alpha/q$ and define the analytic germ*

$$V_{m,\theta}(u; a) = C_a(u)P_m(-m + \theta u, u). \quad (7.17)$$

Then

$$Z_r(-m + \alpha\varepsilon, 1 + \beta\varepsilon; a) = \sum_{h=0}^{r+1} \frac{[u^{r+1-h}]V_{m,\theta}(u; a)}{q^h \varepsilon^h} + \Delta_{m,r+1}(a) + O(\varepsilon). \quad (7.18)$$

In particular,

$$\text{FP}_{\varepsilon=0} Z_r(-m + \alpha\varepsilon, 1 + \beta\varepsilon; a) = [u^{r+1}]V_{m,\theta}(u; a) + \Delta_{m,r+1}(a). \quad (7.19)$$

Every displayed coefficient is a finite expression in rational numbers, a , θ , q^{-1} , and the polygamma values

$$\psi(a), \psi'(a), \dots, \psi^{(r)}(a).$$

Proof. In (7.11), substitute the affine path and put $f(\varepsilon, u) = C_a(u)P_m(-m + \alpha\varepsilon, u)$. The singular term becomes $f(\varepsilon, u)/(q\varepsilon - u)$. For fixed nonzero ε , expand in u first:

$$\frac{1}{q\varepsilon - u} = \sum_{n \geq 0} \frac{u^n}{q^{n+1} \varepsilon^{n+1}}.$$

If $h \geq 1$, collecting the coefficient of $u^r \varepsilon^{-h}$ gives

$$q^{-h} [u^{r+1-h}] C_a(u) P_m \left(-m + \frac{\alpha}{q} u, u \right).$$

For the constant Laurent coefficient the same calculation excludes the term of degree zero in ε in f , and therefore gives

$$[u^{r+1}] C_a(u) (P_m(-m + \theta u, u) - P_m(-m, u)).$$

The holomorphic remainder contributes $[u^r] \mathcal{H}_{m,a}(-m, 1, u)$ to the constant coefficient and nothing to the principal part. Formula (7.14) cancels $-[u^{r+1}] C_a(u) P_m(-m, u)$ and leaves $[u^r] L_m = \Delta_{m,r+1}$. This proves (7.18).

Finally,

$$C_a(u) = \exp \left(- \sum_{k \geq 1} \frac{\psi^{(k-1)}(a)}{k!} u^k \right).$$

Only finitely many coefficients, through degree $r+1$, enter the formula. This proves the stated finite algebra and completes the proof. $\square$

The choice $(\alpha, \beta) = (1, 0)$ recovers the archived common-shift height-one expansion. Indeed comparison of the moving residues in (7.15) proves the finite identity

$$uP_m(-m+u, u) = q_{m+1}(u). \quad (7.20)$$

If the former notation is used, q_{m+1} is precisely its polynomial W_m . The choice $(\alpha, \beta) = (0, 1)$ instead keeps the outer exponent at $-m$ and moves the first inner exponent. Formula (7.19) specifies their discrepancy at every depth; it is not legitimate to transfer a finite part from one direction to the other without that correction.

Corollary 7.4 (Pole order is independent of the transverse direction). *For $m = 0$ or positive odd m , the pole in (7.18) has order exactly $r + 1$, with leading coefficient $\zeta(-m)/q^{r+1}$. For positive even m , it has order exactly r when $r \geq 1$, with leading coefficient*

$$\frac{m+1}{2m} \frac{B_m}{q^r}.$$

At positive even m and $r = 0$ the function is holomorphic at the intersection, and the value is independent of the direction.

Proof. From (7.4), $q_0(0) = 1$ and $q_j(0) = 0$ for $j > 0$. Hence $P_m(-m, 0) = A_{m+1}(-m) = \zeta(-m)$. This proves the first assertion. For positive even m , A_{m+1} is identically zero, so $[u]P_m(-m+\theta u, u)$ is independent of θ . It can be computed at $\theta = 1$ using (7.20).

For completeness, the Gamma-ratio coefficients also obey

$$q_{m+1}(u) = (u-m)_{m+1} [v^{m+1}] e^{-v} \left(\frac{v}{1-e^{-v}} \right)^{1+u}.$$

To justify this auxiliary identity, the beta integral gives, for $\Re u < 0$ and sufficiently large positive x ,

$$\frac{\Gamma(x+u)}{\Gamma(x)} = \frac{1}{\Gamma(-u)} \int_0^\infty e^{-xv} v^{-1-u} e^{-uv} \left(\frac{v}{1-e^{-v}} \right)^{1+u} dv.$$

Expanding the analytic factor at $v = 0$ and integrating a finite Taylor polynomial gives

$$q_j(u) = (-u)_j [v^j] e^{-uv} \left(\frac{v}{1-e^{-v}} \right)^{1+u}.$$

The analytic factor here is obtained from $e^{-v}(v/(1-e^{-v}))^{1+u}$ by replacing v with $-v$. Since $(-1)^j(-u)_j = (u-j+1)_j$, this proves the identity with $j = m+1$. Both sides are polynomials in u , so it holds for every u by polynomial continuation. For even $m \geq 2$, the coefficient at $u = 0$ in the last bracket vanishes, while its coefficient of u is

$$[v^{m+1}] \frac{v}{e^v - 1} \log \frac{v}{1-e^{-v}} = \frac{m+1}{2m} \frac{B_m}{m!}.$$

This follows on multiplying the ordinary Bernoulli series by $v/2 - \sum_{k \geq 1} B_{2k} v^{2k}/(2k(2k)!)$. Since $(u-m)_{m+1} = m!u + O(u^2)$, one obtains $[u^2]q_{m+1} = (m+1)B_m/(2m) \neq 0$. Formula (7.18) now gives the stated order and leading coefficient. For $r = 0$, the local numerator $P_m(s, 0) = A_{m+1}(s)$ vanishes identically, proving joint holomorphy. $\square$

Directions within the polar hyperplane. The condition $q \neq 0$ is essential whenever the pole in Corollary 7.4 is genuine. A nonconstant affine path with $\alpha + \beta = 0$ remains inside $s + t + m - 1 = 0$, so the meromorphic function has no ordinary one-variable restriction along that path. The exception is $r = 0$ and positive even m , where the function is jointly holomorphic: all approach directions, including tangent ones, give

$$Z_0(-m, 1; a) = \Delta_{m,1}(a) + \frac{m+1}{2m} B_m.$$

7.4 Explicit identities and direction corrections

Write $P = \psi(a)$ and $Q = \psi'(a)$. Formula (7.9) gives

$$P_0(\theta u, u) = -\frac{1}{2} + \frac{u}{2} + \frac{\theta - 1}{2}u^2 + \frac{\theta^2 - \theta}{2}u^3 + O(u^4).$$

Consequently,

$$Z_0(\alpha\varepsilon, 1 + \beta\varepsilon; a) = -\frac{1}{2q\varepsilon} + a - \frac{1}{2} + \frac{P}{2} + O(\varepsilon), \quad (7.21)$$

$$Z_1(\alpha\varepsilon, 1 + \beta\varepsilon; a) = -\frac{1}{2q^2\varepsilon^2} + \frac{1+P}{2q\varepsilon} + 1 - a + \frac{\theta - 1}{2} - \frac{P}{2} - \frac{P^2 - Q}{4} + O(\varepsilon). \quad (7.22)$$

Thus at total depth three the finite part changes by exactly $(\theta - \eta)/2$ when the direction parameter changes from η to θ ; the result holds at every common shift a .

At the next outer integer one obtains

$$Z_0(-1 + \alpha\varepsilon, 1 + \beta\varepsilon; a) = -\frac{1}{12q\varepsilon} + \frac{a^2 + a}{4} + \frac{2\theta - 5}{24} + \frac{P}{12} + O(\varepsilon). \quad (7.23)$$

The two coordinate directions differ by $1/12$ in their finite parts. For an example where the highest potential pole cancels, put $L = \gamma + 2\log 2$. At the half shift,

$$Z_1(-2 + \alpha\varepsilon, 1 + \beta\varepsilon; 1/2) = \frac{1}{8q\varepsilon} + \frac{L}{8} - \frac{7 + 12\theta}{288} + O(\varepsilon). \quad (7.24)$$

At $\theta = 1$ this is the previously obtained height-one finite part $L/8 - 19/288$; at $\theta = 0$ it is $L/8 - 7/288$.

In general define the difference between two directional finite parts by $\mathfrak{A}_{m,r}(\theta, \eta; a)$. It has the exact form

$$\mathfrak{A}_{m,r}(\theta, \eta; a) = [u^{r+1}]C_a(u)(P_m(-m + \theta u, u) - P_m(-m + \eta u, u)). \quad (7.25)$$

The Stirling polynomial has canceled completely. At double-zeta depth $r = 0$ this reduces to

$$\mathfrak{A}_{m,0}(\theta, \eta; a) = -(\theta - \eta)\zeta(-m)H_m, \quad H_0 = 0. \quad (7.26)$$

To see the last equality, $[u]$ of the difference is $(\theta - \eta)A'_{m+1}(-m)$. For odd m , $A_{m+1}(s) = B_{m+1}(s)m/(m+1)!$, and its logarithmic derivative at $s = -m$ is $-H_m$; the cases $m = 0$ and positive even m vanish. Thus the double-zeta directional correction is rational and independent of a , even though each individual finite part contains $\psi(a)$.

7.5 An exact normal slice and all Stieltjes derivatives

At depth two one can go beyond the finite part in the direction that keeps the outer exponent fixed. The resulting reduction is a classical Bernoulli specialization of a multiple zeta function; it also supplies particularly short mixed Gamma–Stieltjes identities.

Theorem 7.5 (The complete inner-exponent slice). *For $m \geq 0$, the meromorphic identity*

$$Z_0(-m, t; a) = -\frac{1}{m+1} \sum_{j=0}^{m+1} \binom{m+1}{j} B_j(1)\zeta(t - m - 1 + j, a) \quad (7.27)$$

holds. More generally, for every trailing depth r ,

$$Z_r(-m, t; a) = -\frac{1}{m+1} \sum_{j=0}^{m+1} \binom{m+1}{j} B_j(1)M_r(t - m - 1 + j; a), \quad (7.28)$$

where $M_r(w; a) = \sum_{n \geq 0} e_r(n; a)(n + a)^{-w}$.

Proof. In the half-plane $\Re t > m + 2$, the sum $\sum e_r(n; a)(n + a)^{-t} \zeta(s, n + a + 1)$ is locally normally convergent for s near $-m$. The Bernoulli evaluation

$$\zeta(-m, x + 1) = -\frac{B_{m+1}(x + 1)}{m + 1} = -\frac{1}{m + 1} \sum_{j=0}^{m+1} \binom{m+1}{j} B_j(1) x^{m+1-j}$$

gives the finite formula termwise there. Continuing meromorphically in t proves the identities. This argument specifies the restriction $s = -m$ before taking a Laurent coefficient in t . $\square$

Use the Stieltjes convention

$$\zeta(1 + \varepsilon, a) = \frac{1}{\varepsilon} + \sum_{\ell \geq 0} \frac{(-1)^\ell}{\ell!} \gamma_\ell(a) \varepsilon^\ell, \quad \gamma_0(a) = -\psi(a).$$

Writing $\zeta^{(\ell)}(w, a) = \partial_w^\ell \zeta(w, a)$, (7.27) yields the complete regular coefficients

$$\begin{aligned} [\varepsilon^\ell] Z_0(-m, 1 + \varepsilon; a) = & -\frac{1}{(m+1)\ell!} \sum_{j=0}^m \binom{m+1}{j} B_j(1) \zeta^{(\ell)}(j - m, a) \\ & + \frac{(-1)^\ell}{\ell!} \zeta(-m) \gamma_\ell(a), \quad \ell \geq 0. \end{aligned} \quad (7.29)$$

Its principal part is $\zeta(-m)/\varepsilon$. At $m = 0$ this simplifies to

$$\boxed{\text{FP}_{t=1} \partial_t^\ell Z_0(0, t; a) = -\zeta^{(\ell)}(0, a) + \frac{(-1)^{\ell+1}}{2} \gamma_\ell(a), \quad \ell \geq 0.} \quad (7.30)$$

For $\ell = 1$, Lerch's classical identity [10] gives the concrete logarithmic formula

$$\boxed{\text{FP}_{t=1} \partial_t Z_0(0, t; a) = -\log \Gamma(a) + \frac{1}{2} \log(2\pi) + \frac{1}{2} \gamma_1(a).} \quad (7.31)$$

The logarithm of Γ is its analytic branch on $\Re a > 0$, real for positive a . Equivalently,

$$\begin{aligned} Z_0(0, 1 + \varepsilon; a) = & -\frac{1}{2\varepsilon} + a - \frac{1}{2} + \frac{1}{2} \psi(a) \\ & + \varepsilon \left(-\log \Gamma(a) + \frac{1}{2} \log(2\pi) + \frac{1}{2} \gamma_1(a) \right) + O(\varepsilon^2). \end{aligned} \quad (7.32)$$

This does not evaluate the first regular coefficient in the different slice $Z_0(\varepsilon, 1; a)$; the two slices coincide only at those coefficients explicitly identified above.

There is also a finite antiderivative at every Stieltjes order. Put $J_\ell(a) = \text{FP}_{t=1} \partial_t^\ell Z_0(0, t; a)$. Then

$$\boxed{\int J_\ell(a) da = \frac{\zeta^{(\ell+1)}(0, a)}{2(\ell+1)} - \ell! \sum_{j=0}^{\ell} \frac{\zeta^{(j)}(-1, a)}{j!} + \text{constant}.} \quad (7.33)$$

Indeed the Hurwitz shift derivative implies

$$\partial_a \zeta^{(j)}(-1, a) = \zeta^{(j)}(0, a) - j \zeta^{(j-1)}(0, a), \quad \partial_a \zeta^{(\ell+1)}(0, a) = (-1)^{\ell+1} (\ell+1) \gamma_\ell(a).$$

The finite sum telescopes on differentiation. For example, a primitive of the right side of (7.31) is

$$\frac{1}{4} \zeta''(0, a) - \zeta'(-1, a) - \zeta(-1, a).$$

These formulas provide exact derivative and antiderivative identities in the independent inner spectral direction, including every generalized Stieltjes order.

7.6 Scope, verification, and the next questions

The new finite-part theorem stops at $O(\varepsilon)$ because the first derivatives of $\mathcal{H}_{m,a}$ can enter that term. The normal slice at depth two is exceptional: the Bernoulli reduction evaluates all of its regular coefficients. Formula (7.28) reduces the corresponding problem at larger depth by one, but does not by itself reduce every resulting spectral derivative to ordinary Hurwitz data.

The supplied program `verify_two_exponent.py` checks the residue identity (7.20) exactly for $0 \leq m \leq 6$. It separately evaluates the continued nested sum using its original finite initial segment and a long Euler–Maclaurin tail. Discrete Cauchy averages then extract Laurent coefficients without using the predicted principal part in the extraction. A second cutoff and longer tail check numerical stability. Eight cases at 65 decimal working digits compare 21 Laurent coefficients through the finite part; the maximum absolute discrepancy is below 2.25×10^{-46} . The longer-tail repeat differs by less than 5.9×10^{-52} . These are arbitrary-precision diagnostics; the normal-convergence and coefficient-extraction proofs above establish the identities for all parameters.

Three further research directions are concrete. First, allow the second inner exponent to move as well; this introduces several intersecting polar divisors and tests whether a finite analogue of P_m still eliminates every constant-term remainder. Second, determine linear combinations of first regular coefficients in different transverse directions that cancel the derivatives of $\mathcal{H}_{m,a}$. Third, compare the rational directional correction at $a = 1$ with the general Laurent algorithms and with the Stirling-polynomial formulas for nonpositive multiple-zeta values [16, 17]; this requires matching both the order of indices and the chosen regularization, since the present base point has unit inner exponents.

8 Source audit and proposed corrections

8.1 The Gaussian candidates remain conjectures

The manuscript distinguishes proved reductions from frozen integer-relation candidates. That distinction is preserved here. The current S_6 statement is `cycloquot:conj:S6` in `chapters/04-cyclotomic-quotients.tex`; the current S_8 statement is `s8new:conj:S8` in `chapters/04-S8-candidate.tex` [1]. The latter is a different vector from an earlier rejected S_8 candidate. An objection to that rejected vector is not an objection to the current one.

The supplied rigorous proximity certificates for the current frozen vectors have residual intervals containing zero. In particular, a bound $|R| < 10^{-355}$ proves a very strong approximation but does not prove $R = 0$. Repeating floating-point integer-relation searches would not close that logical gap. Nothing proved in this article supplies the missing exact cyclotomic relation, so neither conjecture is promoted to a theorem.

The finite separating functional already constructed for a restricted level-four quotient is useful but has a limited conclusion. It rules out a derivation inside the particular stated depth-one-product shuffle/stuffle presentation. It does not prove that the target constant lies outside every larger cyclotomic algebra, and it does not rule out octahedral functional equations or additional regularized relations. The exact proof of S_4 in the manuscript, with its explicitly verified additional relation families, is a model for the type of enlargement that would be needed.

8.2 A convergent depth-three bridge for the Gaussian S_6 target

The restricted quotient can be tested against an explicit relation just outside its defining schemas. This gives a concrete exact reduction to include in a future certificate, while leaving the S_6 conjecture open. Here the manuscript’s harmonic sum and Dirichlet beta function are

$$S_p = \sum_{n \geq 0} \frac{(-1)^n H_n}{(2n+1)^p} \quad (p > 1), \quad \beta(s) = \sum_{n \geq 0} \frac{(-1)^n}{(2n+1)^s}.$$

Use the series convention

$$\mathrm{Li}_{s_1, \dots, s_d}(x_1, \dots, x_d) = \sum_{n_1 > \dots > n_d \geq 1} \frac{x_1^{n_1} \cdots x_d^{n_d}}{n_1^{s_1} \cdots n_d^{s_d}}, \quad K_{6,1} = \mathrm{Im} \mathrm{Li}_{6,1}(i, -1).$$

For precision about colors, write $x_j = i^{c_j}$, with c_j read modulo four. With $\omega_a = dt/(t-a)$ and outer-letter-first iterated integrals $\mathcal{I}$, the corresponding word convention is

$$\mathrm{Li}_s(i^{c_1}, \dots, i^{c_d}) = (-1)^d \mathcal{I} \left(0^{s_1-1} i^{-c_1} \dots 0^{s_d-1} i^{-(c_1+\dots+c_d)} \right). \quad (8.1)$$

All words below have first letter different from 1 and last letter different from 0.

Proposition 8.1 (A single-by-double shuffle reduction). *The following identity consists entirely of convergent values:*

$$\begin{aligned} K_{6,1} + \mathrm{Im} \mathrm{Li}_{5,2}(1, -i) &= \sum_{a=1}^5 \mathrm{Im} \mathrm{Li}_{a,6-a,1}(i, -i, -1) \\ &\quad + \mathrm{Im} \mathrm{Li}_{5,1,1}(1, i, i) + \mathrm{Im} \mathrm{Li}_{5,1,1}(1, -1, -i) \\ &\quad - \mathrm{Im} \mathrm{Li}_{1,5,1}(i, 1, -1) - \mathrm{Im} \mathrm{Li}_{5,1,1}(1, i, -1) \\ &\quad - \mathrm{Im} \mathrm{Li}_{5,1,1}(1, -1, i). \end{aligned} \quad (8.2)$$

Proof. Consider $P = \mathrm{Li}_1(i) \mathrm{Li}_{5,1}(1, -1)$. Its stuffle expansion is

$$\begin{aligned} P &= \mathrm{Li}_{1,5,1}(i, 1, -1) + \mathrm{Li}_{5,1,1}(1, i, -1) + \mathrm{Li}_{5,1,1}(1, -1, i) \\ &\quad + \mathrm{Li}_{6,1}(i, -1) + \mathrm{Li}_{5,2}(1, -i). \end{aligned}$$

For its integral shuffle, insert the single letter $-i$ into the word $0^4(1)(-1)$ at each of its seven positions. Formula (8.1) then gives

$$P = \sum_{a=1}^5 \mathrm{Li}_{a,6-a,1}(i, -i, -1) + \mathrm{Li}_{5,1,1}(1, i, i) + \mathrm{Li}_{5,1,1}(1, -1, -i).$$

Subtract the two expansions and take imaginary parts. The first-index-one terms have first argument i and converge by summation by parts; all remaining terms converge absolutely. Thus the finite-sum stuffle limit and the convergent integral shuffle both apply. No divergent $\mathrm{Li}_1(1)$ or regularization parameter is used. $\square$

What this says about the known obstruction. The weight-seven functional ℓ in the pinned manuscript, `cycloquot:eq:separating`, is a functional on a specified formal depth-two presentation, not a functional on actual periods [1]. It annihilates the displayed single-by-single shuffle/stuffle rows and the stated background; in particular,

$$\ell(K_{6,1}) = 1, \quad \ell(\mathrm{Im} \mathrm{Li}_{5,2}(1, -i)) = 0.$$

Extend it to the free higher-depth word space by assigning zero to all depth-three symbols, and call this particular extension $\tilde{\ell}$. For the convergent single-by-double row

$$R = \mathrm{Im}\{\mathrm{shuffle}(\mathrm{Li}_1(i), \mathrm{Li}_{5,1}(1, -1)) - \mathrm{stuffle}(\mathrm{Li}_1(i), \mathrm{Li}_{5,1}(1, -1))\},$$

the explicit expansions prove $\tilde{\ell}(R) = -1$. Hence the naive zero extension of the old separator already fails on this elementary additional row. This calculation does not rule out another extension of ℓ , and it does not establish that any particular additional family, including octahedral relations, is indispensable.

Let Q_7 denote the ten-term right side of (8.2). The ordinary stuffle of $\text{Li}_5(1) \text{Li}_2(-i)$ gives

$$\text{Im Li}_{5,2}(1, -i) = g_{2,5} + \beta(7) - G\zeta(5), \quad g_{a,b} = \text{Im Li}_{a,b}(i, 1), \quad G = \beta(2).$$

Selecting the even inner indices gives $S_6 = g_{6,1} + K_{6,1}$, so there is the exact reduction

$$S_6 = g_{6,1} - g_{2,5} + G\zeta(5) - \beta(7) + Q_7. \quad (8.3)$$

The unresolved task is to reduce this specified Q_7 to the frozen conjectural basket by a finite exact certificate. The double-shuffle identities used here are classical; the contribution of this audit is the explicit target row and the precise diagnostic of the inherited obstruction.

The accompanying `verify_s6_target.py` independently enumerates the seven integral shuffles and five stuffle terms using exact integer arithmetic. It also checks all 192 defining weight-seven restricted rows against ℓ , verifies the value -1 above, and records the unchanged conjectural target in `s6_target_audit.json`.

8.3 Three carry-forward manuscript corrections

The following points were already identified in the incoming Twisted Stieltjes report [5]. They remain relevant at the pinned revision. The accompanying patch proposes only these narrow changes and is supplied unapplied.

A real logarithm for the negative-parameter integral. In `chapters/03-algebraic.tex`, the displayed formula labelled `cleo:eq:lintanh` uses $\log \tan(\theta/2)$ while its stated real parameter range includes negative values. The real formula needs $\log |\tan(\theta/2)|$. Under the principal complex logarithm, the incorrect right side at $\lambda = -1/2$ differs from the real integral by $-i\pi^2/6$. This is a branch correction, with the absolute value fixed by the real integral and its real derivative.

Remove an unsupported nonreduction description. The same chapter calls a specific exact diagonal value *non-elementary* without defining a comparison algebra or proving non-membership. Keep the exact value and remove that classification. An available special-function representation by itself proves no nonreduction theorem.

Scope the coefficient field to the character in use. In `chapters/07-integration.tex`, the field $\mathbb{Q}(\sqrt{q})$ belongs to the particular quadratic-character example. General additive or multiplicative character coordinates can require cyclotomic and character values. The patch makes this scope explicit without altering the quadratic calculation.

One earlier proposed wording change is already integrated: the PSLQ discussion now allows known dependencies to be removed or quotiented out and requires a nonzero coefficient of the target. It no longer requires a prior linear-independence theorem. That edit is therefore not repeated or counted as a new correction.

8.4 Corrections in the inspected Borwein–Dilcher author PDF

The author-hosted PDF of Borwein and Dilcher, *Derivatives and fast evaluation of the Witten zeta function*, <https://carmamaths.org/resources/jon/omega3.pdf>, contains two transcription errors relevant to a Mellin implementation. Lemma 2.1(b), printed page 2, assigns $H_0 = 1$; the required convention is $H_0 = 0$. In Theorem 2.6, printed page 5, the positive-integer formula for B_p has an extra factor H_{p-1} . The corrected coefficients are

$$A_p = \frac{(-1)^{p-1} H_{p-1}}{\Gamma(p)}, \quad B_p = \frac{(-1)^p}{\Gamma(p)}, \quad p = 1, 2, \dots \quad (8.4)$$

These findings concern the inspected PDF; they do not assert that every published version repeats them.

Here is an independent check. Coalescing the Gamma pole and the $\zeta(1)$ term in Jonquière's formula gives

$$\mathrm{Li}_p(e^{-x}) = \sum_{\substack{k \geq 0 \\ k \neq p-1}} \zeta(p-k) \frac{(-x)^k}{k!} + \frac{(-x)^{p-1}}{(p-1)!} (H_{p-1} - \log x), \quad 0 < x < 2\pi. \quad (8.5)$$

Thus the coefficient multiplying $x^{p-1} \log x$ is precisely B_p in (8.4). At $p = 3$, the required coefficient is $-1/2$, whereas the printed B_3 is $-3/4$. Separately, $\mathrm{Li}_1(e^{-x}) = -\log(1 - e^{-x}) = -\log x + O(x)$ forces $H_0 = 0$. A test at $p = 2$ would miss the B_p error because $H_1 = 1$.

Theorems 2.5 and 2.6 also state $\theta > 0$ for their local power-series expansions. For ordinary convergent summation, impose $0 < \theta < 2\pi$, as required by Jonquière's local series [11, 25.12.12]. A larger splitting point is possible only with an additional continuation or intermediate-integral prescription.

To see the issue without an error estimate, take $r = s = 0, t = 3$ in the double local series and keep $v = 0, u = 2k - 1$. The absolute value of this summand is

$$\frac{|B_{2k}| \theta^{2k+2}}{2(2k)!(2k+2)} = \frac{\zeta(2k) \theta^2}{2k+2} \left(\frac{\theta}{2\pi} \right)^{2k}.$$

It fails to tend to zero for $\theta > 2\pi$. The original Witten sum at $(0, 0, 3)$ converges, so the obstruction belongs to the displayed local series representation. The corrected implementation can retain the classical theorem by choosing any $0 < \theta < 2\pi$; it does not need a new assignment to divergent power series.

8.5 Further transcription corrections in a Bailey–Borwein author PDF

The following corrections concern the inspected author PDF of *Computation and theory of Mordell–Tornheim–Witten sums II*, dated 5 May 2014, printed pages 17–19 [8]. Equation numbers in this subsection refer to that PDF. No assertion about other versions is intended. Use its derivative convention

$$\begin{aligned} \omega_{a,b,c}(q, r, s) &= \sum_{n,m \geq 1} \frac{(-\log n)^a (-\log m)^b (-\log(n+m))^c}{n^q m^r (n+m)^s}, \\ \zeta_{a,b}(r, s) &= \sum_{k > j \geq 1} \frac{(-\log k)^a (-\log j)^b}{k^r j^s}. \end{aligned} \quad (8.6)$$

Superscripts on Li and ζ below denote derivatives with respect to their orders; a, b, c are nonnegative integers.

Order-derivative signs in Eq. (84). The factors printed as $(-1)^n \log^a n$ and $(-1)^m \log^b m$ must be $(-1)^a \log^a n$ and $(-1)^b \log^b m$. Writing $\Gamma(s, z) = \int_z^\infty e^{-y} y^{s-1} dy$, the corrected formula is

$$\begin{aligned} \omega_{a,b,c}(q, r, s) &= \sum_{n,m \geq 1} \frac{(-\log n)^a (-\log m)^b}{n^q m^r} \partial_s^c \left\{ \frac{\Gamma(s, n+m)}{\Gamma(s)(n+m)^s} \right\} \\ &\quad + \int_{1/e}^1 \partial_s^c \left\{ \frac{(-\log u)^{s-1}}{\Gamma(s)} \right\} \mathrm{Li}_q^{(a)}(u) \mathrm{Li}_r^{(b)}(u) \frac{du}{u}. \end{aligned} \quad (8.7)$$

Indeed $\partial_q^a n^{-q} = (-\log n)^a n^{-q}$. Expanding the two polylogarithms in the integral from 0 to $1/e$ and setting $y = -\log u$ gives the incomplete-Gamma term, including the complete derivative of its Gamma ratio. There is no alternation in n or m .

The sign in Eq. (86). The corrected identity is

$$\omega_{0,0,b}(0,0,t) = \zeta^{(b)}(t-1) - \zeta^{(b)}(t). \quad (8.8)$$

For $t > 2$, exactly $k-1$ positive pairs have $n+m=k$, so its left side is $\sum_{k \geq 2} (k-1)(-\log k)^b/k^t$. This proves (8.8) directly. The printed right side has the opposite sign; already at $b=0$ it is negative while the defining sum is positive.

Swapping orders also swaps derivative labels in Eqs. (90)–(91). The corrected formulas are

$$\omega_{a,0,b}(s,0,t) + \omega_{b,0,a}(t,0,s) = \zeta^{(a)}(s)\zeta^{(b)}(t) - \zeta^{(a+b)}(s+t), \quad (8.9)$$

$$\omega_{a,b,0}(s,t,0) - \omega_{b,0,a}(t,0,s) - \omega_{a,0,b}(s,0,t) = \zeta^{(a+b)}(s+t). \quad (8.10)$$

To see the indexing, $\omega_{a,0,b}(s,0,t) = \zeta_{b,a}(t,s)$. Differentiate $\zeta(t,s) + \zeta(s,t) = \zeta(s)\zeta(t) - \zeta(s+t)$ a times in s and b times in t . The second omega term must therefore carry the derivative labels $(b,0,a)$ when its spectral arguments are $(t,0,s)$. The correction is invisible in the special case $a=b$.

Gamma arguments in Eqs. (93)–(94). Gamma functions and their derivatives are evaluated at the spectral variable, not the number of differentiations. The corrected ladders are

$$\begin{aligned} \zeta_{b,a}(t,s) = & - \sum_{k=0}^{b-1} \binom{b}{k} \frac{\Gamma^{(b-k)}(t)}{\Gamma(t)} \zeta_{k,a}(t,s) \\ & + \frac{1}{\Gamma(t)} \int_0^1 \frac{\text{Li}_s^{(a)}(x)}{1-x} \log^b(-\log x) (-\log x)^{t-1} dx, \end{aligned} \quad (8.11)$$

$$\begin{aligned} \omega_{a,b,c}(q,r,s) = & - \sum_{k=0}^{c-1} \binom{c}{k} \frac{\Gamma^{(c-k)}(s)}{\Gamma(s)} \omega_{a,b,k}(q,r,s) \\ & + \frac{1}{\Gamma(s)} \int_0^1 \text{Li}_q^{(a)}(x) \text{Li}_r^{(b)}(x) \log^c(-\log x) (-\log x)^{s-1} \frac{dx}{x}. \end{aligned} \quad (8.12)$$

An empty sum is zero. For (8.11), differentiate

$$\Gamma(t)\zeta_{0,a}(t,s) = \int_0^1 (-\log x)^{t-1} \frac{\text{Li}_s^{(a)}(x)}{1-x} dx$$

b times with respect to t and isolate the term with no derivative on $\Gamma(t)$. The same Leibniz argument applied to $\Gamma(s)\omega_{a,b,0}(q,r,s)$ gives (8.12). Thus the occurrences of the Gamma argument b in Eq. (93) must be t , and those of c in Eq. (94) must be s .

Direct convergence domains. These derivations require actual convergence before any continuation. For real $q, r \geq 0$, the integral formulas (8.7) and (8.12) hold under

$$s > 0, \quad q + s > 1, \quad r + s > 1, \quad q + r + s > 2. \quad (8.13)$$

The printed total-weight condition alone omits the two pair conditions: at $(q,r,s) = (2,0,1)$ the $n=1$ subseries is harmonic and diverges. Splitting the sum into $m \leq n$ and $n < m$ proves sufficiency of the three strict summability conditions; finite logarithmic powers do not alter this open domain. Formula (8.11) holds, for example, when $t > 1$ and $t+s > 2$, while (8.9)–(8.10) follow directly for $s, t > 1$. The latter identities and (8.8) then extend meromorphically. Such continuation does not assert ordinary convergence of the displayed integrals or series outside their stated domains.

8.6 Conventions that must survive integration

No new false theorem was found among the inspected periodic Hurwitz, coordinate-residue, and coincident-pair premises used in this article. The following assumptions are nevertheless part of the mathematics and must remain attached to the statements.

1. The pointwise product finite part, the periodic distributional convolution, and the canonical extension in the shift coordinate are different constructions. Theorem 3.1 computes their relation, including its contact term.
2. The argument finite part uses a specified cutoff coordinate. In Theorem 2.1, subtracting only the residue restricted to a polar hyperplane loses finite numerator derivatives and changes the requested answer.
3. A fixed-integer identity may be differentiated only in variables that remain free in that identity. In particular, the finite positive-integer Tornheim sums in (5.2) do not determine derivatives in either fixed inner order.
4. An origin ray is restricted before its removable singularity is filled in. The two distinct values $1/3$ and $5/12$ are both correct for their respective rays.
5. Numerical verification checks an implementation and can detect mistakes. It does not replace the proofs or establish arithmetic independence.

The source audit and correction patch are separate from the new mathematical section files. Their provenance notes identify which findings are inherited, which are newly checked against an external PDF, and which are restrictions on interpretation.

9 Further research questions

The formulas above suggest concrete next problems. The questions below are proposals, not assertions of currently unproved identities. They are ordered roughly by how directly the present proofs provide the next calculation.

9.1 Cubic and higher coincident products

1. Reduce the third diagonal Tornheim derivative. Theorem 2.3 isolates $\omega'''(0)$ as the remaining global coordinate for the cubic digamma finite part. Seek an exact reduction of this derivative to a stated algebra of ordinary zeta derivatives, Gamma or Barnes values, and convergent logarithmic integrals. A useful first target is a second independent functional identity for the cubic integral, rather than an unqualified integer-relation search in a large constant basket. If no such reduction is found, an explicit minimal integral representation is still valuable, but it must not be called an independence result.

2. Determine every asymmetric cubic ray derivative. In the local decomposition (6.9), the next Taylor layer involves $H_{AA} = H_{BB}$, H_{AB} , $H_{AC} = H_{BC}$, and H_{CC} . The exact C axis determines H_{CC} . The diagonal cubic coefficient supplies one linear combination of the other three. Can two additional exact restrictions or functional equations determine the remainder? A concrete deliverable would be a formula for $W'''_{a,b,c}(0)$ with its smallest explicit collection of convergent-integral coordinates. This would clarify exactly where ordinary single-zeta data cease to determine all directions.

3. Give the explicit global kernel at degree four. The local subtraction at any number of factors is already finite: (2.4) lists the needed Taylor coefficients. The unsolved part is the arithmetic of the global Fourier zero-frequency constraint. At degree four, separate the sign partitions of $n_1 + n_2 + n_3 + n_4 = 0$ and identify a finite set of constrained multiple sums whose meromorphic continuations represent all moments. The first useful test is the fully coincident FP $\int_0^1 \psi(x)^4 dx$ with every counterterm written out. This would extend the present cubic kernel without assuming single-zeta closure.

4. Analyze triple collisions on unequal grids. The incoming pair formulas allow unequal spatial dilations. A triple product introduces several collision loci and a possible full three-way coincidence. Derive a chamber formula in the shift variables, identify the exact changes on crossing each collision wall, and compare the successive pair-collision limits with the direct triple-coincident finite part. The challenge is to preserve one stated coordinate prescription throughout; repeated finite-part operations need not commute automatically.

9.2 Contact distributions and coordinate transport

5. Extend the contact generator to several shifts. Theorem 3.1 determines the entire point-supported correction for two factors and one shift. For a three-factor correlation, the collision sets are lines and points in a two-shift torus. Determine the distributions supported on those sets and their intersections. The pairwise terms should recover (3.10); the new information is the correction at a joint intersection. A meaningful answer should specify test-function normalizations and not introduce a product of singular distributions without a definition.

6. Characterize invariant products exactly. Corollary 4.4 gives a clean sufficient criterion: at most one zero Stieltjes index. It is not a complete classification of products or linear combinations of products. Apply Theorem 4.1 to the finite Taylor jets of the analytic factors and identify all cancellations among the forbidden monomials. One concrete starting point is the span of $\gamma_i \gamma_j \gamma_k$ at fixed total index, with coefficients in a prescribed constant field. The field matters because a functional cancellation and an arithmetic relation are different questions.

7. Determine the coordinate-defect composition law. For two tangent coordinates f and g , express the defect for $f \circ g$ in terms of the defects for f and g and the corresponding pullbacks. Identify a finite representation of this law on pole order at most R and logarithmic degree at most L . The invariant subspace and its exact codimension are now known. The remaining structural question is the action on the quotient, including whether a natural coordinate choice gives a canonical representative for each obstruction class.

9.3 Harmonic Laurent data and primitives

8. Move all the harmonic exponents independently. Theorem 7.3 treats an outer exponent and the first inner exponent while every trailing exponent remains one. At the first new case $\zeta_a(s, t_1, t_2)$, perturb all three orders independently. Find the complete set of local polar divisors, then determine which principal coefficients and finite parts admit finite Gamma, Bernoulli, and Stirling formulas without an unevaluated holomorphic remainder. General multiple-zeta Laurent algorithms already exist [16]; the target is a compact explicit specialization that preserves the common shift and all directions.

9. Classify directions with explicit higher regular terms. The two-exponent theorem stops at the finite part for a reason: derivatives of its holomorphic remainder generally enter later. The exact normal slice in Theorem 7.5 is an exception with an all-order Stieltjes formula.

Determine other integer outer-order slices or direction combinations for which the remainder cancels to a prescribed order. A finite criterion would organize which higher coefficients genuinely require new multiple zeta derivatives.

10. Normalize primitives using Gamma multiplication. Equation (7.33) supplies a finite primitive in the shift variable, and (3.17) supplies mean-zero periodic primitives. Study their compatibility with reflection, multiplication of the shift, and rational shifts. At low derivative orders, compare the resulting constants with standard Barnes multiple-Gamma normalizations. Any claimed simplification should specify both the additive constant and the branch of the logarithm.

9.4 Colored sums and exact Gaussian certificates

11. Reduce mixed inner signs at arbitrary depth. The rational theorem settles the sector with all inner orders nonpositive and arbitrary positive integer weights. Positive integer orders at depth two are handled by partial fractions. Develop a finite nested-polylogarithm reduction for arbitrary depth with both positive and nonpositive inner orders, while retaining the full complex outer order. Coincident colors and repeated poles must be built into the formula. An affine shift alone is a routine Lerch or Hurwitz extension and is less informative than this mixed sector.

12. Detect arithmetic cancellation of depth-two jets. At roots of unity, (5.19) and (5.20) completely describe the depth-one terms. Which character projections, symmetrizations, or distribution combinations cancel the residual depth-two order derivatives in (5.7)? A useful result would be an exact finite linear map whose kernel identifies combinations reducible to Gamma, Hurwitz-zeta, and generalized-Stieltjes data. Such a map should retain the allowed derivative directions instead of specializing before differentiation.

13. Seek exact S_6 and S_8 certificates. Use the frozen target vectors at the pinned revision and enlarge the proved relation space, not the numerical precision alone. The existing quotient obstruction identifies relations that cannot suffice; the verified S_4 octahedral proof identifies additional families worth testing. For every added regularized relation, prove the regulator prescription and retain its exact rational coefficient row. A successful certificate must sum to the target identity exactly. If a restricted search fails, publish the precisely defined relation space and its separating functional, without extending the impossibility statement beyond that space.

14. Formalize the finite proof kernels. The coefficient extractions in the subset completion, contact generator, and harmonic residue polynomial are finite algebra. Separate that algebra from the analytic lemmas of normal convergence and meromorphic continuation, and formalize each part with explicit interfaces. In particular, a reusable proof of the Taylor-subtraction lemma in several spectral variables would prevent accidental interchange of a coordinate finite part and a spectral Laurent constant in future integrations.

10 Verification and reproducibility

The proofs establish the all-parameter identities. The accompanying code checks finite algebra, signs, specializations, and independent analytic representations. Exact rational checks are distinguished from arbitrary-precision floating-point diagnostics throughout. The latter do not provide interval-certified error bounds.

10.1 Independent algebraic checks

The coordinate-invariance script compares two constructions of the defect: a residue extraction and the binomial coefficient formula. It performs 140 such exact comparisons, checks 36 obstruction ranks, and verifies 440 Stieltjes threshold and sharp-coefficient instances. All are finite symbolic computations.

The spectral script performs 402 exact checks: positive-inner-order partial fractions, composition polynomials, harmonic–Stirling coefficients, rational partial-fraction reconstructions, independently convolved coefficient sequences, and the explicit weighted cyclotomic certificate. The harmonic script verifies recovery of the inherited one-direction residue formula for $m = 0, \dots, 6$. An independent symbolic expansion of the local Tornheim decomposition gives identically zero residuals for the constant, linear, and quadratic coefficients in arbitrary a, b, c .

For contacts, the direct A -generator and the symmetric Gamma–cosine form (3.7) agree through total degree four, after the exact even-zeta relations are applied. The script also checks all seven displayed low-index coefficients and the swap/reflection identity through total argument order five.

The focused S_6 audit independently regenerates the convergent single-by-double row in Proposition 8.1, checks all 192 rows of the restricted presentation against its separator, and verifies the exact target normalization and two admissible Cayley seeds. These checks certify the stated reduction and the obstruction diagnostic; they do not certify the conjecture.

10.2 Numerical checks using different representations

Identity family	Independent comparison	Largest observed absolute discrepancy
Cubic digamma finite part	Local convergent digamma integral versus a Jonquière/incomplete-Gamma evaluation of $\omega'''(0)$	2.19×10^{-35}
Multivariate cubic completion	Direct Taylor-subtracted Hurwitz integral versus three independently continued Tornheim kernels; real and complex asymmetric triples	5.84×10^{-50}
Distributional contact generator	Full meromorphic Hurwitz Fourier coefficient versus canonical off-point coefficient plus contact; 189 mode comparisons	2.71×10^{-76}
Quadratic Tornheim rays	Independent Mellin continuation and symmetric extrapolation on four distinct rays	2.60×10^{-37}
Weighted and root-of-unity identities	Nine convergent Mellin quadratures versus Gamma/zeta specializations	4.68×10^{-61}
Two-exponent harmonic Laurent data	21 coefficients in eight cases, extracted by discrete Cauchy averages from a continued nested sum	2.25×10^{-46}
Stieltjes primitive hierarchy	Six numerical shift derivatives compared with generalized Stieltjes values	2.04×10^{-56}

The cubic calculation uses 65 decimal working digits and refines its finite-difference step. The leading known diagonal derivatives are recovered well beyond the precision required for the cubic comparison. The multivariate complex case was repeated with longer independent

truncations; its cross-error decreased to 5.22×10^{-58} , while the direct integral changed by less than 1.51×10^{-63} .

The contact check uses 85 working digits, three asymmetric complex parameter pairs, frequencies $-3, 1, 4$, and all $p + q \leq 5$. Its maximum scaled residual is less than 2.52×10^{-84} . Omitting the proposed contact term in a negative control produces an error of approximately 3.2457, so the comparison is sensitive to the delta contribution. Zero modes and reflection signs are checked separately.

The harmonic extraction uses 65 working digits and keeps the first 64 original summands, with 32 Euler–Maclaurin tail terms and 32-point Cauchy averages on radius 0.01. It does not subtract the predicted Laurent expression before extraction. Repeating one evaluation with cutoff 88 and 40 tail terms changes the result by less than 5.9×10^{-52} . Details and parameters are preserved in the JSON outputs.

10.3 Package organization and integration

The archive contains the modular `article.tex`, a flattened standalone $\text{\TeX}$ source, the compiled PDF, section sources, reproducible Python scripts, their data outputs, and a source receipt. It also includes a theorem-status ledger, independent review notes, an integration guide, and the unapplied correction patch. The `README.md` gives the build and verification commands. The scripts depend on Python, SymPy, and mpmath; the document uses standard pdf $\text{\LaTeX}$ packages and requires no external bibliography processor.

For manuscript integration, retain the proof of the cubic local completion before the asymmetric ray theorem, because the ray proof uses the diagonal quadratic coefficient. The contact and coordinate classification sections can be integrated independently once their finite-part conventions are retained. The spectral and harmonic sections use independent notation and state the free variables needed for their derivative claims.

The source receipt pins this continuation to a concrete repository state. The patch was checked against that state and remains unapplied. No theorem status in the repository is changed by the existence of this archive; the result ledger is a proposal for integration with the corresponding proofs.

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- [3] ProveIt incoming report, *Exact Resonance and Coordinate Transport*, 10 October 2026, archive `ProveIt_Resonance_Coordinate_Transport_2026-10-10.zip`. [Pinned archive](#). Used for coordinate transport and the normal spectral-derivative question.
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